

X



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bhargavi111gm@gmail.com ✓

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » The Joy of Computing using Python (course)



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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

- Introduction to Programming (unit? unit=17&lesson=18)
- Why Programming? (unit?

Week 1 : Assignment 1

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

Assignment submitted on 2025-02-05, 22:01 IST

1) Which blocks can be used to move the cat? 1 point

- ☐ Walk
- ☒ Move
- ☐ Teleport
- ☒ Glide

Yes, the answer is correct.

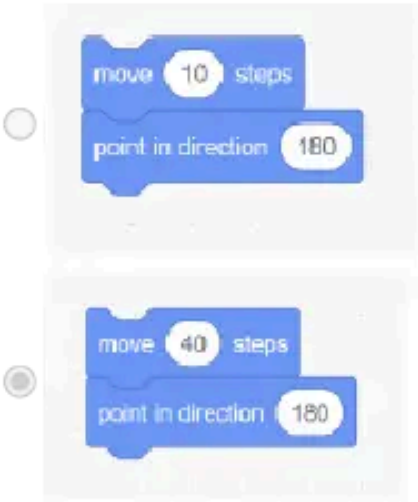
Score: 1

Accepted Answers:

Move

Glide

2) Which set of blocks moves the cat 40 steps and then points down ? 1 point



unit=17&lesso
n=19)

• Programming
for Everybody
(unit?
unit=17&lesso
n=20)

• Any
Prerequisites?
(unit?
unit=17&lesso
n=21)

• Where to
start? (unit?
unit=17&lesso
n=22)

• Why do we
have so many
languages?
(unit?
unit=17&lesso
n=23)

• How to go
about
programming?
(unit?
unit=17&lesso
n=24)

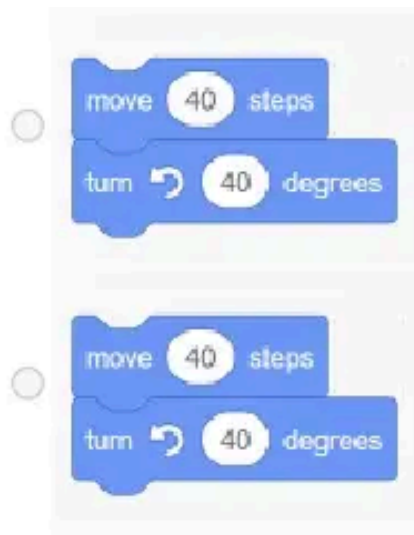
• Why to learn
programming?
(unit?
unit=17&lesso
n=25)

• What is
programming?
(unit?
unit=17&lesso
n=26)

• How to give
instructions?
(unit?
unit=17&lesso
n=27)

• Introduction to
Scratch (unit?
unit=17&lesso
n=28)

• Introduction to
Loops (unit?



Yes, the answer is correct.
Score: 1

Accepted Answers:



3) What does the cat say here ?

1 point



- ☐ Says the number 7, for 1 second, 7 times and changes the value of number variable by multiplying previous value with 1.
- ☐ Says the number 7, for 1 second, 7 times and changes the value of number variable by adding previous value with 1.
- ☒ Says the number 7, for 1 second, then says numbers from 6 to 1 one by one for 1 second each.
- ☐ Says the number 7, for 1 second, then says numbers from 6 to 0 one by one for 1 second each.

Yes, the answer is correct.

unit=17&less
n=29)

☐ More about
Loops (unit?
unit=17&less
n=30)

☐ Solution to
Looping
Problem (unit?
unit=17&less
n=31)

☐ Scratch :
Animation 1
(unit?
unit=17&less
n=32)

☐ Scratch :
Animation 2
(unit?
unit=17&less
n=33)

☐ Scratch :
Animation 3
(unit?
unit=17&less
n=34)

☐ More on
Scratch (unit?
unit=17&less
n=35)

☒ Week 1
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=17&less
n=36)

☒ Quiz: Week 1
: Assignment
1
(assessment?
name=501)

Week 2 ()

Week 3 ()

week 4 ()

Score: 1

Accepted Answers:

Says the number 7, for 1 second, then says numbers from 6 to 1 one by one for 1 second each.

4) From the previous question, what is the value of myNum variable after the loop ends ? **1 point**

- ☐ 1
☐ 2
☐ 7
☒ 0

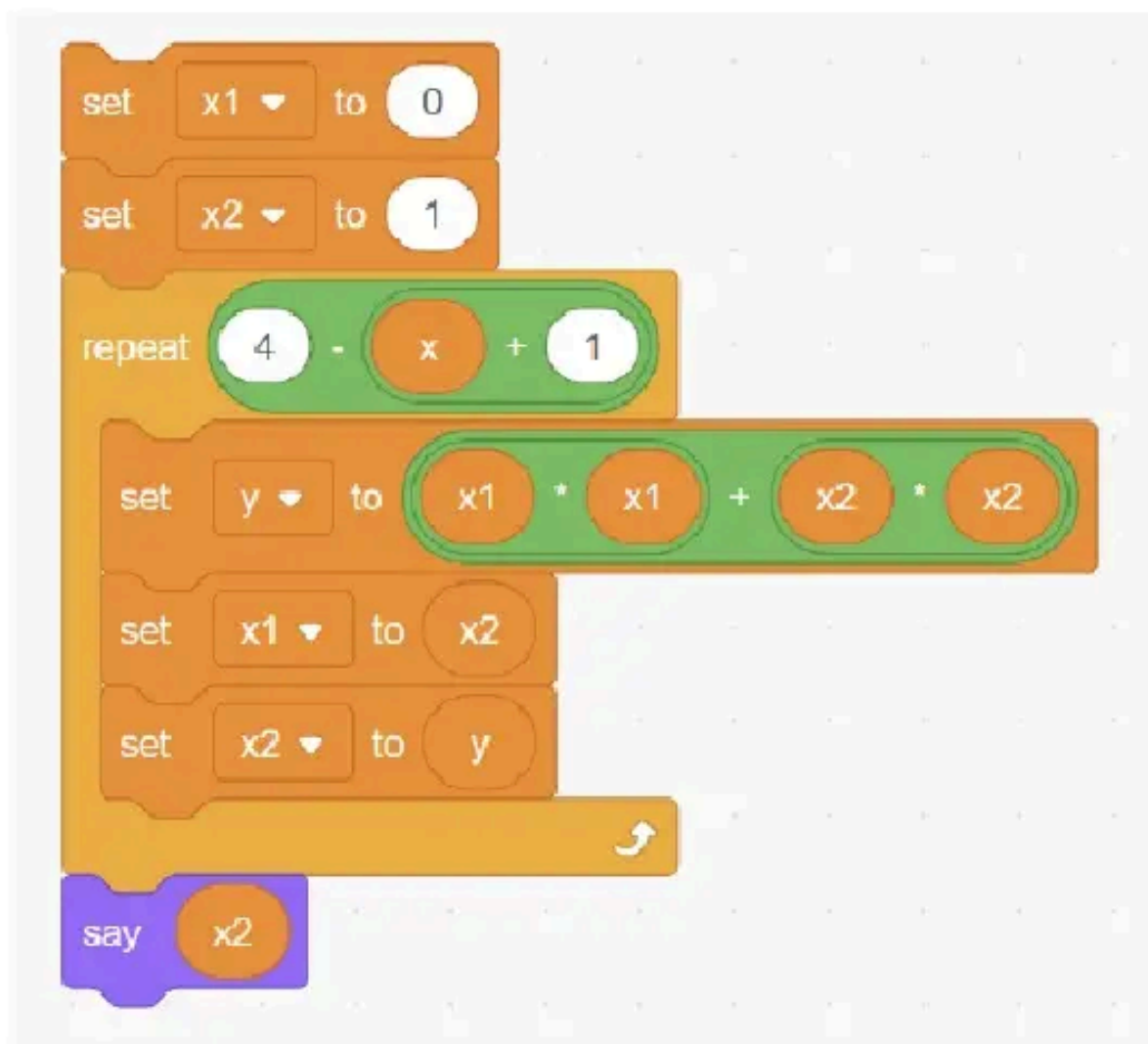
Yes, the answer is correct.

Score: 1

Accepted Answers:

0

5) Let answer for previous question be stored in variable x, what does the cat say after **1 point** the execution of the loop ?



- ☐ 16
☒ 5
☐ 29
☐ 44

Yes, the answer is correct.

Week 5 ()**Week 6 ()****Week 7 ()****Week 8 ()****Week 9 ()****Week 10 ()****Week 11 ()****Week 12 ()****Text
Transcripts
()****Download
Videos ()****Books ()****Problem
Solving
Session -
Jan 2025 ()**

Score: 1

Accepted Answers:

5

6) Which of the following data types is primarily used for storing a sequence of characters in most programming languages?

1 point

- ☐ Integer
- ☐ Float
- ☒ String
- ☐ Boolean

Yes, the answer is correct.

Score: 1

Accepted Answers:

String

7) Which of the following statements best describes why loops are commonly used ? **1 point**

- ☐ Loops make the code run slower but more reliably.
- ☒ Loops allow you to execute a block of code repeatedly without having to write it multiple times.
- ☐ Loops are only useful for error handling.
- ☐ Loops automatically convert variables from one data type to another.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Loops allow you to execute a block of code repeatedly without having to write it multiple times.

8) Suppose you have an arithmetic expression involving addition, subtraction, multiplication, and exponentiation. Which statement is true about instructing a computer to evaluate it? **1 point**

- ☐ Such expressions cannot be computed.
- ☐ A computer can only evaluate expressions involving addition and multiplication.
- ☒ These expressions can be evaluated using built-in operations or libraries.
- ☐ A computer can only evaluate expressions involving addition and subtraction.

Yes, the answer is correct.

Score: 1

Accepted Answers:

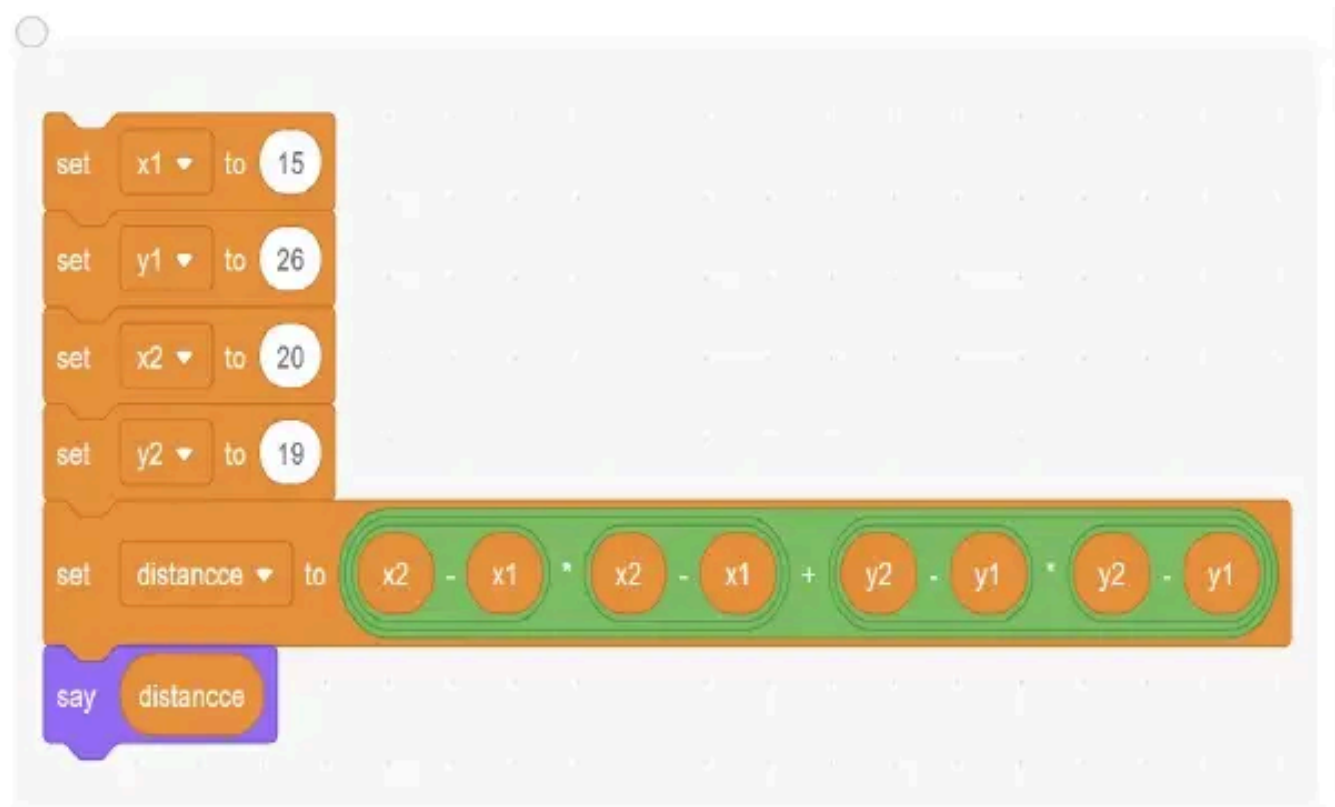
These expressions can be evaluated using built-in operations or libraries.

9) In which of the following code blocks **calculation** of the squared distance between two points (15, 25) and (20, 19) is accomplished. **1 point**

The image displays three sequential screenshots of a Scratch script, illustrating the calculation of the distance between two points. The script is composed of the following blocks:

- Four 'set' blocks for variables: $x1$ to 15, $y1$ to 25, $x2$ to 20, and $y2$ to 19.
- A 'set' block for 'distance' to the expression: $(x2 - x1)^2 + (y2 - y1)^2$.
- A 'say' block with the text 'distance'.

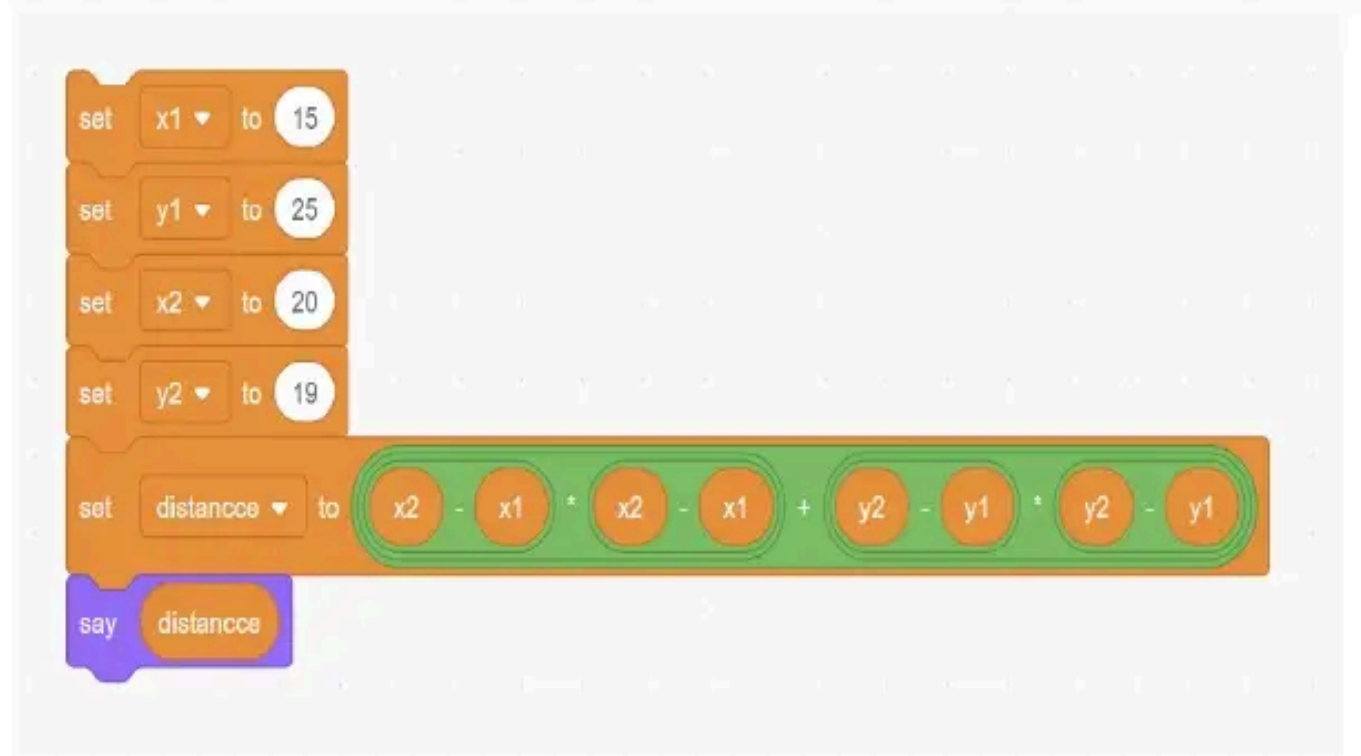
The three screenshots show the state of the script at different points during execution, with the values of the variables changing as the script progresses. In the first screenshot, $x1=15, y1=25, x2=20, y2=19$. In the second, $x1=15, y1=20, x2=19, y2=25$. In the third, $x1=15, y1=20, x2=19, y2=26$.



Yes, the answer is correct.

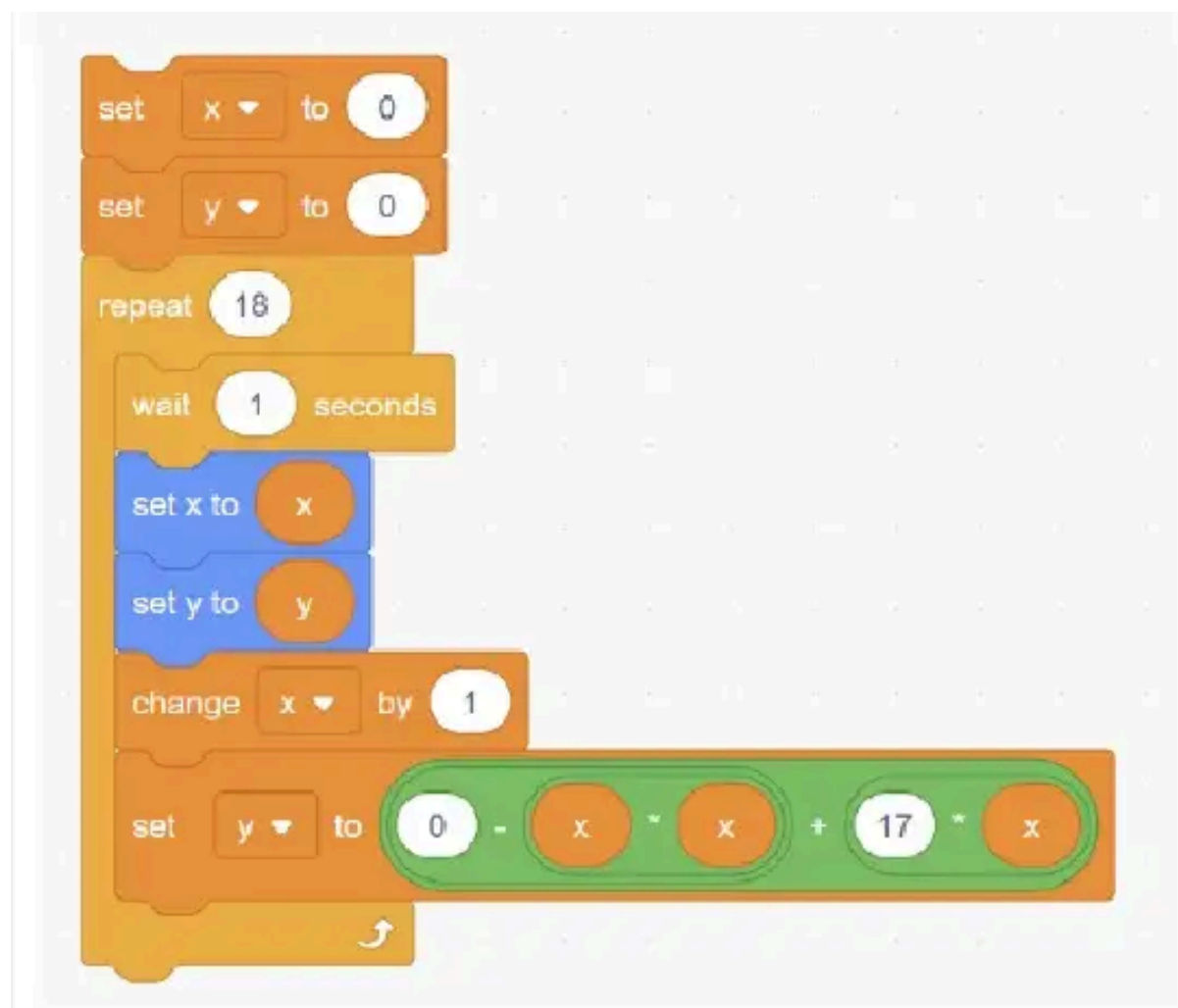
Score: 1

Accepted Answers:



10) What task does the block below accomplish ?

1 point



- ☐ Moves the ball in a parabolic path and stops 18 away steps from center(0, 0) in x direction.
- ☐ Moves the ball in a circular path and 18 times.
- ☐ Moves the ball in a parabolic path and stops 18 away steps from center(0, 0) in y direction.
- ☒ Moves the ball in a parabolic path and stops 17 away steps from center(0, 0) in x direction.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Moves the ball in a parabolic path and stops 18 away steps from center(0, 0) in x direction.

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

- ☐ Introduction to
Anaconda
(unit?
unit=37&lesso
n=38)

- ☐ Installation of
Anaconda
(unit?

Week 2 : Assignment 2

The due date for submitting this assignment has passed.

Due on 2025-02-05, 23:59 IST.

Assignment submitted on 2025-02-05, 22:02 IST

1) **Statement:** If a variable is assigned a value once, the value in the variable cannot be changed in the variable. **1 point**

- ☐ False, the variable stores all values it was assigned.
- ☒ False, the variable stores the value from the latest assignment.
- ☐ True, the variable stores only the value from the second-last assignment.
- ☐ True, the variable stores value from the initial assignment only.

Yes, the answer is correct.

Score: 1

Accepted Answers:

False, the variable stores the value from the latest assignment.

2) Which of the following code blocks print - "Ramesh talks to Suresh and Kamlesh" ? **1 point**

- ☒

```
print("Ramesh talks to Suresh and Kamlesh")
```
- ☐

```
print("Ramesh talks to " + "Suresh" + " and", "Kamlesh")
```
- ☒

```
print("Ramesh talks to Suresh", "and", "Kamlesh")
```
- ☐

```
print("Ramesh talks to" + "Suresh and Kamlesh")
```

Partially Correct.

Score: 0.67

Accepted Answers:

```
print("Ramesh talks to Suresh and Kamlesh")
```

unit=37&less
n=39)

Introduction to
Spyder IDE
(unit?
unit=37&less
n=40)

Printing
statements in
Python (unit?
unit=37&less
n=41)

Understanding
Variables in
Python (unit?
unit=37&less
n=42)

Executing a
sequence of
instructions in
the Console
(unit?
unit=37&less
n=43)

Writing your
First Program
(unit?
unit=37&less
n=44)

Taking inputs
from the user
(unit?
unit=37&less
n=45)

Discount
Calculation
(unit?
unit=37&less
n=46)

Motivation to if
condition
(unit?
unit=37&less
n=47)

A reminder on
how to deal
with numbers
(unit?
unit=37&less
n=48)

```
print("Ramesh talks to " + "Suresh" + " and", "Kamlesh")
print("Ramesh talks to Suresh", "and", "Kamlesh")
```

3) What aren't the correct ways to inform python that input is a decimal point number ? **1 point**

- ☒ in(input())
- ☐ float(input())
- ☒ int(input())
- ☒ a = input()
a = int(a)

Yes, the answer is correct.

Score: 1

Accepted Answers:

in(input())

int(input())

a = input()

a = int(a)

4) The following program outputs 723 -

0 points

```
def mystery(container):
    result = []
    for i in range(len(container)):
        if i % 2 != 0:
            result.append(container[i] * 4)
        else:
            result.append(container[i] * (-1))
    return result
```

For what value of **a** does the code output 123 ?

- ☐ 1
- ☐ 2
- ☒ 6
- ☐ 5

Yes, the answer is correct.

Score: 0

Accepted Answers:

6

5) What does previous question calculate ?

0 points

- ☐ Calculates the factorial of **a**.
- ☒ Calculates the factorial of **a-1** and adds 3
- ☐ Calculates the factorial of **a-1+3**.
- ☐ Calculates the **a** multiples of **a** starting from 1 and adds 3.

Yes, the answer is correct.

Score: 0

Accepted Answers:

*Calculates the factorial of **a-1** and adds 3*

☐ Understanding if condition's working (unit? unit=37&less n=49)

☐ Realizing the importance of syntax and indentation (unit? unit=37&less n=50)

☐ Introductions to loops (unit? unit=37&less n=51)

☐ Loops: Sum of numbers (unit? unit=37&less n=52)

☐ Loops: Sum of numbers (continued) (unit? unit=37&less n=53)

☐ Loops: Multiplication Tables (unit? unit=37&less n=54)

☐ Introduction to While Loop (unit? unit=37&less n=55)

☐ Week 2 Feedback Form: The Joy of Computing using Python (unit? unit=37&less n=56)

☒ **Quiz: Week 2 : Assignment 2 (assessment? name=503)**

☒ Week 2: Programming

6) Which loop is used to perform a set of repetitive tasks without a condition by default **1 point** in Python?

- ☐ while loop
- ☒ for loop
- ☐ do-while loop
- ☐ while-range loop

Yes, the answer is correct.

Score: 1

Accepted Answers:

for loop

7) What happens when the condition inside the **if** and **while** evaluate to false ? **1 point**

- ☐ Python interpreter ignores the **if/while** blocks, and halts the program.
- ☒ Python interpreter ignores the **if/while** blocks, and proceeds the program from the lines after the **if/while** block.
- ☐ Python interpreter executes the **if/while** blocks, and rest of the program.
- ☐ Python interpreter executes the **if/while**, and the programs runs in an infinite loop.

Yes, the answer is correct.

Score: 1

Accepted Answers:

*Python interpreter ignores the **if/while** blocks, and proceeds the program from the lines after the **if/while** block.*

8) The following program might/might not have an infinite loop. Does the program have infinite loop ? **0 points**

```
n = ?
k = 0
a = []
b = 0
while (n != 0):
    k = k + (n % 10)
    a.append(n % 10)
    n = n // 10
a.sort()
for i in a:
    b = b + i
if(b != k):
    print("PYTHON IS FUN")
else:
    print("PYTHON IS BORING")
```

- ☐ No, the program doesn't have infinite loop.
- ☒ Yes, it can be prevented by updating the value of **a** before the **if** block at line 3.
- ☐ Yes, it can be prevented by removing both the **if** blocks inside the while loop.
- ☐ Yes, but it cannot be prevented.

Yes, the answer is correct.

Score: 0

Accepted Answers:

*Yes, it can be prevented by updating the value of **a** before the **if** block at line 3.*

Assignment 1
(/noc25_cs69/
progassignment?name=504)

Week 2:
Programming
Assignment 2
(/noc25_cs69/
progassignment?name=505)

Week 2:
Programming
Assignment 3
(/noc25_cs69/
progassignment?name=506)

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

Text
Transcripts
()

Download
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Books ()

Problem
Solving
Session -
Jan 2025 ()

9) For which of the following values of **name** and **age** variables does the following code print "You are lucky"?

1 point

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
flag = "False"
if age >= 18:
    flag = "True"
else:
    flag = "False"
counter = 0
for i in name:
    if (i == "a"):
        counter += 1
if (flag == "True"):
    if (counter >= 2):
        print("You are lucky")
else:
    print("You are not lucky")
```

- ☒ aryan, 20
☐ arjun, 19
☐ aakash, 16
☒ anand, 18

Yes, the answer is correct.

Score: 1

Accepted Answers:

aryan, 20

anand, 18

10) For which of the options among the previous question, the program doesn't print anything.

1 point

- ☐ aryan, 20
☒ aakash, 16
☐ arjun, 19
☐ anand, 18

Yes, the answer is correct.

Score: 1

Accepted Answers:

aakash, 16

X



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Course
outline

About NPTEL
()

How does an
NPTEL online
course work?
()

Week 1 ()

Week 2 ()

Week 3 ()

☒ Lists Part 1 :
Introduction
(unit?
unit=57&lesson=
58)

☐ Lists Part 2 :
Manipulation
(unit?
unit=57&lesson=
59)

Week 3 : Assignment 3

The due date for submitting this assignment has passed.

Due on 2025-02-12, 23:59 IST.

Assignment submitted on 2025-02-12, 21:37 IST

1) Which of the following is/are **false** statement(s)?

1 point

- ☒ Lists are fixed in size, they cannot store more than 1024 values
- ☐ One can access element in list by using non-numeric indices
- ☐ Iterating over lists is possible in Python
- ☒ We need to specify required size of list while creating a new list variable

Partially Correct.
Score: 0.67

Accepted Answers:
Lists are fixed in size, they cannot store more than 1024 values
One can access element in list by using non-numeric indices
We need to specify required size of list while creating a new list variable

2) In the below code -

○ Lists Part 3 :
Operations
(unit?
unit=57&lesson=
60)

○ Lists Part 4 :
Slicing (unit?
unit=57&lesson=
61)

○ Loops and
Conditionals :
Fizzbuzz 01
(unit?
unit=57&lesson=
62)

○ Loops and
Conditionals :
Fizzbuzz 02
(unit?
unit=57&lesson=
63)

○ Crowd
Computing - Just
estimate 01
(unit?
unit=57&lesson=
64)

○ Crowd
Computing - Just
estimate 02
(unit?
unit=57&lesson=
65)

○ Crowd
Computing - Just
estimate 03
(unit?
unit=57&lesson=
66)

○ Crowd
Computing - Just
estimate 04
(unit?
unit=57&lesson=
67)

○ Crowd
Computing - Just
estimate 05
(unit?
unit=57&lesson=
68)

○ Crowd
Computing - Just
estimate 06
(unit?

```
n= ?
a = []
for i in range(n):
    l1 = []
    p = 0
    for j in range(i):
        l1.append(j)
    for k in l1:
        p+=1
    a.append(p)
k=0
for l in a:
    k+=1
print(k)
```

For what value of **n** does the program print **10** ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

3) From the previous question, for what values of **n** is the number **25** appended to list a.

1 point

- ☒ 25
☐ 24
☒ 35
☒ 32

No, the answer is incorrect.

Score: 0

Accepted Answers:

35

32

4) What does the following code perform ?

1 point

```
def mystery(container):
    result = []
    for i in range(len(container)):
        if i % 2 != 0:
            result.append(container[i] * 4)
        else:
            result.append(container[i] * (-1))
    return result
```

- It converts any input list into a new list which is filled with some numbers which are always in increasing order
☒ It converts any input list into a new list such that at even indices, the value is a multiple of an even number and at odd indices, the value is either even or negative number

unit=57&lesson=69)

Permutations - Jumbled Words 01 (unit? unit=57&lesson=70)

Permutations - Jumbled Words 02 (unit? unit=57&lesson=71)

Permutations - Jumbled Words 03 (unit? unit=57&lesson=72)

Theory of Evolution 01 (unit? unit=57&lesson=73)

Theory of Evolution 02 (unit? unit=57&lesson=74)

Theory of Evolution 03 (unit? unit=57&lesson=75)

Theory of Evolution 04 (unit? unit=57&lesson=76)

Week 3 Feedback Form : The Joy of Computing using Python!! (unit? unit=57&lesson=77)

Quiz: Week 3 : Assignment 3 (assessment? name=508)

Week 3: Programming Assignment 1 (/noc25_cs69/pr ogassignment? name=510)

- ☐ It converts any input list into a new list such that at odd indices, the value is always even number and at even indices, the value is negative
- ☐ It converts any input list into a new list, which follows no pattern

No, the answer is incorrect.
Score: 0

Accepted Answers:

It converts any input list into a new list such that at odd indices, the value is always even number and at even indices, the value is negative

5) From the previous question, if the option -

1 point

It converts any input list into a new list such that at even indices, the value is a multiple of an even number and at odd indices, the value is either even or negative number.

is incorrect, Can you make changes to code such that it this option is always true for any input ?

- ☐ No, it is not possible to make such changes
- ☒ No, the option is already correct
- ☐ Yes, we can make changes

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes, we can make changes

6) If **file.txt** exists, Does the code successfully run ?

1 point

```
with open("file.txt", "r") as f:
    data = f.read()
    print(data)
```

- ☐ Yes
- ☒ No

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes

7) Which of the following are examples of Social Computing ?

1 point

- ☒ ChatGPT
- ☒ Wikipedia
- ☒ Quora
- ☒ Twitch

No, the answer is incorrect.
Score: 0

Accepted Answers:

Wikipedia

Quora

8) What does the following code print for **n = 256**?

1 point

Week 3: Programming Assignment 2 (/noc25_cs69/progassignment?name=511)
Week 3: Programming Assignment 3 (/noc25_cs69/progassignment?name=513)
week 4 ()
Week 5 ()
Week 6 ()
Week 7 ()
Week 8 ()
Week 9 ()
Week 10 ()
Week 11 ()
Week 12 ()
Text Transcripts ()
Download Videos ()
Books ()
Problem Solving Session - Jan 2025 ()

```
n = ?
k = 0
a = []
b = 0
while (n != 0):
    k = k + (n % 10)
    a.append(n % 10)
    n = n // 10
a.sort()
for i in a:
    b = b + i
if(b != k):
    print("PYTHON IS FUN")
else:
    print("PYTHON IS BORING")
```

- ☒ PYTHON IS BORING
- ☐ PYTHON IS FUN

Yes, the answer is correct.
Score: 1
Accepted Answers:
PYTHON IS BORING

9) From the previous question, is **a.sort()** responsible for printing of either of the two possible sentences ? **1 point**

- ☐ Yes, and the code may print "PYTHON IS FUN"
- ☒ No, it is purely dependent on variable n
- ☐ No, it is not dependent on a.sort(), and the code will never print "PYTHON IS FUN"

No, the answer is incorrect.
Score: 0
Accepted Answers:
No, it is not dependent on a.sort(), and the code will never print "PYTHON IS FUN"

10) What does the code in question 8, calculate ?
Can you say what the values of k and b are if $n = 245326785456 \times 10^{3359}$.
If values of k and b are different, enter 0, else enter value of k.

64

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 57

1 point

X



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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

☐ Practice is the
key (unit?)

Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-02-19, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following options provides the general formula for the magic constant **1 point** of a magic square of size n , where all elements are distinct numbers from 1 to n^2 ?

☐
$$\frac{n(n^2 + 1)}{2}$$

☐
$$\frac{n^3}{2}$$

☐
$$\frac{n^3 + 2n}{2}$$

☐
$$\frac{n^4 + n^2}{2n}$$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$$\frac{n(n^2 + 1)}{2}$$

$$\frac{n^4 + n^2}{2n}$$

2) What would the magic constant be for a magic square of size 7, given that all **1 point** elements in the square are distinct numbers from 1 to 49?

- ☐ 260
- ☐ 111
- ☐ 175

unit=78&less
n=79)

☐ Magic Square:
Hit and Trial
01 (unit?
unit=78&less
n=80)

☐ Magic Square:
Hit and Trial
02 (unit?
unit=78&less
n=81)

☐ Magic Square:
Hit and Trial
03 (unit?
unit=78&less
n=82)

☐ Magic Square:
Hit and Trial
04 (unit?
unit=78&less
n=83)

☐ Magic Square:
Hit and Trial
05 (unit?
unit=78&less
n=84)

☐ Let's program
and play (unit?
unit=78&less
n=85)

☐ Dobble Game
- Spot the
similarity 01
(unit?
unit=78&less
n=86)

☐ Dobble Game
- Spot the
similarity 02
(unit?
unit=78&less
n=87)

☐ Dobble Game
- Spot the
similarity 03
(unit?
unit=78&less
n=88)

☐ 165

No, the answer is incorrect.

Score: 0

Accepted Answers:
175

3) Does transposing any magic square change the sum across some rows/columns/diagonal?

1 point

☐ Yes

☐ No

No, the answer is incorrect.

Score: 0

Accepted Answers:
No

4) Which of the following are valid magic squares ?

1 point

☐
$$\begin{bmatrix} 10 & 3 & 13 & 8 \\ 5 & 16 & 2 & 11 \\ 4 & 9 & 7 & 14 \\ 15 & 6 & 12 & 1 \end{bmatrix}$$

☐
$$\begin{bmatrix} 20-e & 6-e & 26-e & 16-e \\ 10-e & 32-e & 4-e & 22-e \\ 8-e & 18-e & 14-e & 28-2e \\ 30-e & 12-e & 24-e & 2-e \end{bmatrix}$$

☐
$$\begin{bmatrix} 1 & 14 & 4 & 15 \\ 8 & 11 & 5 & 10 \\ 13 & 2 & 16 & 3 \\ 12 & 7 & 9 & 6 \end{bmatrix}$$

☐
$$\begin{bmatrix} \pi & 14\pi & 4\pi & 15\pi \\ 8\pi & 11\pi & 5\pi & 10\pi \\ 13\pi & 2\pi & 16\pi & 3\pi \\ 12\pi & 7\pi & 9\pi & 6\pi \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 10 & 3 & 13 & 8 \\ 5 & 16 & 2 & 11 \\ 4 & 9 & 7 & 14 \\ 15 & 6 & 12 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 14 & 4 & 15 \\ 8 & 11 & 5 & 10 \\ 13 & 2 & 16 & 3 \\ 12 & 7 & 9 & 6 \end{bmatrix}$$

☐ Dobble Game
- Spot the
similarity 04
(unit?
unit=78&less
n=89)

☐ What is your
date of birth?
(unit?
unit=78&less
n=90)

☐ Birthday
Paradox - Find
your twin 01
(unit?
unit=78&less
n=91)

☐ Birthday
Paradox - Find
your twin 02
(unit?
unit=78&less
n=92)

☐ Birthday
Paradox - Find
your twin 03
(unit?
unit=78&less
n=93)

☐ Birthday
Paradox - Find
your twin 04
(unit?
unit=78&less
n=94)

☐ Birthday
Paradox - Find
your twin 05
(unit?
unit=78&less
n=95)

☐ What's your
favourite
movie? (unit?
unit=78&less
n=96)

☐ Guess the
Movie Name
01 (unit?
unit=78&less
n=97)

$$\begin{bmatrix} \pi & 14\pi & 4\pi & 15\pi \\ 8\pi & 11\pi & 5\pi & 10\pi \\ 13\pi & 2\pi & 16\pi & 3\pi \\ 12\pi & 7\pi & 9\pi & 6\pi \end{bmatrix}$$

5) What is the minimum number of people required to ensure that at least three of them share the same 30-minute birth interval? The intervals start from 12:00 AM and each interval lasts for half an hour.

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 97

1 point

6) Calculate the magic constant for 5x5 square, where all elements are distinct numbers from 1 to 25, is it same as the magic constant for Ramanujan's magic square ?

If yes, enter 0, else enter the absolute difference between the two.

Hint: Search the about Ramanujan's magic square.

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 74

1 point

7) What task does **mystery1()** perform?

1 point

```
def mystery1(number):
    list1 = []
    for i in range(1, number):
        if number % i == 0:
            list1.append(i)
    return list1

def mystery2(n1,n2):
    flag = False
    list2 = []
    for i in mystery1(n1):
        for j in mystery1(n2):
            if i == j:
                flag = True
                list2.append(i)
    if len(list2) > 1:
        print("Completed")
```

☐ Guess the Movie Name 02 (unit? unit=78&lesso n=98)

☐ Guess the Movie Name 03 (unit? unit=78&lesso n=99)

☐ Guess the Movie Name 04 (unit? unit=78&lesso n=100)

☐ Guess the Movie Name 05 (unit? unit=78&lesso n=101)

☐ Guess the Movie Name 06 (unit? unit=78&lesso n=102)

☐ Week 4 Feedback Form: The Joy of Computing using Python (unit? unit=78&lesso n=103)

☐ **Quiz: Week 4 : Assignment 4 (assessment? name=515)**

☒ Week 4: Programming Assignment 1 (/noc25_cs69/progassignment?name=518)

☒ Week 4: Programming Assignment 2 (/noc25_cs69/progassignment?name=516)

- ☐ Calculate factorial of number n.
- ☐ Calculate factors of number n.
- ☐ Calculate factors of number n+1 excluding n.
- ☐ Calculate factors of number n excluding n.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Calculate factors of number n excluding n.

8) For what **n1**, **n2** will the variable **flag** inside **mystery2()** be equal to **True**?

1 point

- ☐ 1,2
- ☐ 2,3
- ☐ 3,4
- ☐ 0,0

No, the answer is incorrect.
Score: 0

Accepted Answers:

2,3

3,4

9) If all possible pairs of prime numbers between 0 and 10, are given to n1 and n2, for **1 point** how many pairs would 2 print "Completed" ?

- ☐ It will print "Completed" only for pairs (2,3)(7,2),(2,5), and for the remaining it would not print "Completed".
- ☐ It will print "Completed" only for pair (2,3), and for the remaining other pairs of primes it would not print "Completed".
- ☐ It will print "Completed" for all pairs of primes between 0 and 20.
- ☐ It will not print "Completed" for any pair.

No, the answer is incorrect.
Score: 0

Accepted Answers:

It will not print "Completed" for any pair.

10) If numbers of pairs of primes which result in mystery2 to print "Completed" are **1 point** lesser than 1, Can we edit the code in **mystery2()** so that "Completed" is always printed for any pair of primes ?

- ☐ Yes, we can change the logic for setting **flag** variable to False.
- ☐ No, it is logically not possible.
- ☐ Yes, we can change/decrease the threshold for length of **list2** in the last **if** block.
- ☐ Yes, we can change the initial value of flag to False, instead of True.

No, the answer is incorrect.
Score: 0

Accepted Answers:

*Yes, we can change/decrease the threshold for length of **list2** in the last **if** block.*

 Week 4:
Programming
Assignment 3
(/noc25_cs69/
progassignment?
name=517)

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 5 : Assignment 5

The due date for submitting this assignment has passed.
Due on 2025-02-26, 23:59 IST.

Assignment submitted on 2025-02-26, 23:14 IST

1) Which of the following is the correct way to add data with key as **CS102** and value **1 point** as **"Database Management Systems"** to a dictionary named **courses** ?

- ☐ `courses["CS101"] = ("Database Management Systems")`
- ☐ `courses["CS102"]["Database Management Systems"]`
- ☒ `courses["CS102"] = "Database Management Systems"`
- ☐ `courses["CS101"] = "Database Management Systems"`

Yes, the answer is correct.
Score: 1
Accepted Answers:
`courses["CS102"] = "Database Management Systems"`

2) What is the probability of Monty not opening the door with goat, given the hypothesis that you initially chose the door which has car ? 1 point

- ☒ 0
- ☐ 0.5
- ☐ 1
- ☐ 0.33

Yes, the answer is correct.
Score: 1
Accepted Answers:
0

- ☐ Introduction to Dictionaries (unit? unit=104&lesson=105)
- ☐ Speech to Text : No need to write 01 (unit? unit=104&lesson=106)
- ☐ Speech to Text : No need to write 02 (unit? unit=104&lesson=107)
- ☐ Speech to Text : No need to write 03 (unit? unit=104&lesson=108)
- ☐ Monte Hall : 3 doors and a twist 01 (unit? unit=104&lesson=109)
- ☐ Monte Hall : 3 doors and a twist 02 (unit? unit=104&lesson=110)
- ☐ Rock, Paper and Scissor : Cheating not allowed !! 01 (unit? unit=104&lesson=111)
- ☐ Rock, Paper and Scissor : Cheating not allowed !! 02 (unit? unit=104&lesson=112)
- ☐ Rock, Paper and Scissor : Cheating not allowed !! 03 (unit? unit=104&lesson=113)

3) What should be replaced with ? in line 10, so that there is high chance that **final_choice** is equal to 2 ?

1 point

```
import random
initial_choice = random.randint(0, 2)
doors = ['goat', 'goat', 'car']

for i in range(3):
    if i != initial_choice and doors[i] != 'car':
        monty_opens = i
        break

if ? :
    for i in range(3):
        if i != initial_choice and i != monty_opens:
            final_choice = i
            break
else:
    final_choice = initial_choice
```

- ☒ 2
- ☐ $2 * (1024) \% 2 \neq 0$
- ☒ $2 * (2 * (89) + 7) \% 2 == 0$
- ☐ $(2 * (90)) \% \text{len}(\text{doors}) \neq 1$

Yes, the answer is correct.

Score: 1

Accepted Answers:

2
 $2 * (2 * (89) + 7) \% 2 == 0$

4) Given that you have a sorted list of 2048 elements, what is the maximum number of comparisons to search such an element using linear search ? Can binary search do it in less number of comparisons ? **1 point**

- ☐ Yes, binary search can do it more efficiently, linear search takes 1024 comparisons
- ☒ Yes, binary search can do it more efficiently, linear search takes 2048 comparisons
- ☐ No, binary search may/may not do it more efficiently, linear search takes 1024 comparisons
- ☐ No, binary search will take more comparisons, linear search takes 2048 comparisons

Yes, the answer is correct.

Score: 1

Accepted Answers:

Yes, binary search can do it more efficiently, linear search takes 2048 comparisons

5) What is the primary use of a file with a .wav or .wave extension?

1 point

- ☐ Storing video recordings
- ☐ Storing log files
- ☐ Storing waves of files
- ☒ Storing audio recordings

- ☐ Rock, Paper and Scissor : Cheating not allowed !! 04 (unit? unit=104&lesson=114)
- ☐ Sorting and Searching : 20 questions game 01 (unit? unit=104&lesson=115)
- ☐ Sorting and Searching : 20 questions game 02 (unit? unit=104&lesson=116)
- ☐ Sorting and Searching : 20 questions game 03 (unit? unit=104&lesson=117)
- ☐ Sorting and Searching : 20 questions game 04 (unit? unit=104&lesson=118)
- ☐ Sorting and Searching : 20 questions game 05 (unit? unit=104&lesson=119)
- ☐ Sorting and Searching : 20 questions game 06 (unit? unit=104&lesson=120)
- ☐ Sorting and Searching : 20 questions game 07 (unit? unit=104&lesson=121)
- ☐ Sorting and Searching : 20

Yes, the answer is correct.

Score: 1

Accepted Answers:

Storing audio recordings

6) What does this program print in the end ?

1 point

```
import random
n = 10
counter = 0
for i in range(n):
    choices = ["rock", "paper", "scissors"]
    choice1 = random.choice(choices)
    choice2 = random.choice(choices)
    if choice1 != choice2:
        counter += 1
print(counter/n)
```

- ☒ Fraction of throws where both players showed different symbol(rock/paper/scissors)
- ☐ Fraction of throws where both players showed rock.
- ☐ Fraction of throws where both players showed same symbol(rock/paper/scissors)
- ☐ Fraction of throws where both players showed paper.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Fraction of throws where both players showed different symbol(rock/paper/scissors)

7) In binary search the list gets divided in every iteration to find the element.

1 point

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

8) Which of the following are requirements for binary search to work correctly?

1 point

- ☐ The list can be unsorted.
- ☐ The list must contain only positive integers.
- ☒ The list must be sorted.
- ☐ The list must have no duplicate values.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The list must be sorted.

9) Given an array [4, 2, 7, 1, 3], what will be the array after the third pass of Bubble Sort?

1 point



questions
game 08 (unit?
unit=104&less
on=122)

☐ Week 5
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=104&less
on=123)

☒ **Quiz: Week 5
: Assignment
5
(assessment?
name=519)**

☒ Week 5:
Programming
Assignment 1
(/noc25_cs69/
progassignme
nt?name=520)

☒ Week 5:
Programming
Assignment 2
(/noc25_cs69/
progassignme
nt?name=521)

☒ Week 5:
Programming
Assignment 3
(/noc25_cs69/
progassignme
nt?name=522)

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

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Transcripts**

☐ [1, 3, 2, 4, 7]
☐ [2, 4, 1, 3, 7]
☒ [1, 2, 3, 4, 7]
☐ [2, 1, 3, 4, 7]

Yes, the answer is correct.
Score: 1
Accepted Answers:
[1, 2, 3, 4, 7]

10) How many swaps are performed in iteration 3 for bubble sorting list [5,4,3,2,1]? Enter 0 if the list is sorted in less than 3 iterations.

2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2

1 point

()

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outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 6 : Assignment 6

The due date for submitting this assignment has passed.

Due on 2025-03-05, 23:59 IST.

Assignment submitted on 2025-03-04, 12:21 IST

1) If n is a positive integer, what is the output of the function given input n , **1 point**

```
def mystery(n):  
    if n <= 0:  
        return 0  
    return n + mystery(n - 1)
```

- ☒ Sum of numbers from 1 to n
- ☐ Sum of numbers from 1 to $n-1$
- ☐ $n-1$
- ☐ n

Yes, the answer is correct.

Score: 1

Accepted Answers:

Sum of numbers from 1 to n

2) **Assertion (A):** If a recursive function doesn't have a base case, it may never stop. **1 point**
Reason (R): Recursion means that a function calls itself.

- ☐ Both (A) and (R) are true, and (R) is the correct explanation of (A).
- ☒ Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- ☐ (A) is true, but (R) is false.
- ☐ (A) is false, but (R) is true.

☐ Substitution Cipher -The science of secrecy (unit? unit=124&less on=125)

☐ Substitution Cipher -The science of secrecy 01 (unit? unit=124&less on=126)

☐ Substitution Cipher -The science of secrecy 02 (unit? unit=124&less on=127)

☐ Substitution Cipher -The science of secrecy 03 (unit? unit=124&less on=128)

☐ Tic Tac Toe - Down the memory Lane (unit? unit=124&less on=129)

☐ Tic Tac Toe - Down the memory Lane 01 (unit? unit=124&less on=130)

☐ Tic Tac Toe - Down the memory Lane 02 (unit? unit=124&less on=131)

☐ Tic Tac Toe - Down the memory Lane 03 (unit? unit=124&less on=132)

Yes, the answer is correct.

Score: 1

Accepted Answers:

Both (A) and (R) are true, but (R) is not the correct explanation of (A).

3) What is the output of following code ?

1 point

```
def func(x):
    return x ** 2

def func(x, y=1):
    return x**2 + y

print(func(5))
```

☐ Error

☐ 25

☒ 26

☐ 5

Yes, the answer is correct.

Score: 1

Accepted Answers:

26

4) In English, the letter 'z' usually appears far less often than most letters. Now imagine applying a Substitution Cipher to a large body of English text, where 'z' is replaced by 'r', and every other letter is also mapped to a unique letter. After encryption, which letter is likely to be the least frequent in the new text?

1 point

Hint: Search the internet for more info, if needed.

☐ z

☒ r

☐ e

☐ It is not possible to determine

Yes, the answer is correct.

Score: 1

Accepted Answers:

r

5) If you have a large ciphertext produced by a simple substitution cipher on an English text, can you determine the original message by examining how often each letter appears?

1 point

Hint: Search the internet for more info, if needed.

☒ Yes, it can be done.

☐ Tic Tac Toe -
Down the
memory Lane
04 (unit?
unit=124&less
on=133)

☐ Tic Tac Toe -
Down the
memory Lane
05 (unit?
unit=124&less
on=134)

☐ Recursion
(unit?
unit=124&less
on=135)

☐ Recursion 01
(unit?
unit=124&less
on=136)

☐ Recursion 02
(unit?
unit=124&less
on=137)

☐ Recursion 03
(unit?
unit=124&less
on=138)

☐ Recursion 04
(unit?
unit=124&less
on=139)

☐ Recursion 05
(unit?
unit=124&less
on=140)

☐ Recursion 06
(unit?
unit=124&less
on=141)

☐ Week 6
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=124&less
on=142)

☒ **Quiz: Week 6
: Assignment**

☐ No, it cannot be done.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Yes, it can be done.

6) Suppose you suspect that a piece of ciphertext was produced using a simple sub- **1 point**
stitution cipher. Under which circumstances might frequency analysis fail to yield accurate
results?

☒ The ciphertext is short.

☒ The ciphertext was encrypted with another method before encrypting with sub-stitution
cipher.

☐ The ciphertext is very long.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The ciphertext is short.

The ciphertext was encrypted with another method before encrypting with sub-stitution cipher.

7) If variable **dict_name** is a non-empty dictionary, what does dict name.values() re- **1 point**
turn?

☐ Returns nothing, but prints all the keys in the dictionary.

☐ Returns a list of all the keys in the dictionary.

☒ Returns a list of all the values in the dictionary.

☐ Returns a list of all the items in the dictionary.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Returns a list of all the values in the dictionary.

8) **Assertion (A):** A Caesar Cipher is a type of Substitution Cipher. **1 point**
Reason (R): In a Caesar Cipher, each letter in the plaintext is replaced by a letter at a fixed shift
in the alphabet.

☒ Both (A) and (R) are true, and (R) is the correct explanation of (A).

☐ Both (A) and (R) are true, but (R) is not the correct explanation of (A).

☐ (A) is true, but (R) is false.

☐ (A) is false, but (R) is true.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Both (A) and (R) are true, and (R) is the correct explanation of (A).

9) Which outcome generally follows from providing a proper base case in a recursive **1 point**
function?

☒ The function returns a predefined result when the base condition is met.

☒ It prevents infinite recursion and stops when a specific condition is met.

6
(assessment?
name=523)

- Week 6:
Programming
Assignment 1
(/noc25_cs69/
progassignme
nt?name=524)
- Week 6:
Programming
Assignment 2
(/noc25_cs69/
progassignme
nt?name=525)
- Week 6:
Programming
Assignment 3
(/noc25_cs69/
progassignme
nt?name=526)

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

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Problem
Solving
Session -
Jan 2025 ()

- ☐ It ensures recursion depth is always constant.
- ☐ It allows the function to keep calling itself indefinitely.

Yes, the answer is correct.
Score: 1
Accepted Answers:
The function returns a predefined result when the base condition is met.
It prevents infinite recursion and stops when a specific condition is met.

10) In a standard 3×3 Tic-Tac-Toe board, how many distinct winning lines (rows, columns, or diagonals) pass through the center square?

1 point

- ☐ 3
- ☐ 2
- ☐ 5
- ☒ 4

Yes, the answer is correct.
Score: 1
Accepted Answers:
4

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About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 7 : Assignment 7

The due date for submitting this assignment has passed.
Due on 2025-03-12, 23:59 IST.

Assignment submitted on 2025-03-12, 20:56 IST

1) What will be the output of the following code? 1 point

```
my_dict = {1: 2, 2: 3, 3: 4}
x = my_dict[my_dict[1]]
print(x)
```

- ☐ 2
- ☒ 3
- ☐ 4
- ☐ Error

Yes, the answer is correct.
Score: 1
Accepted Answers:
3

2) What is the spiral order traversal of the given matrix? 1 point

Week 7 ()

- ☐ Snakes and Ladders - Not on the Board (unit? unit=143&less on=144)
- ☐ Snakes and Ladders - Not on the Board - Part 01 (unit? unit=143&less on=145)
- ☐ Snakes and Ladders - Not on the Board - Part 02 (unit? unit=143&less on=146)
- ☐ Snakes and Ladders - Not on the Board - Part 03 (unit? unit=143&less on=147)
- ☐ Snakes and Ladders - Not on the Board - Part 04 (unit? unit=143&less on=148)
- ☐ Snakes and Ladders - Not on the Board - Part 05 (unit? unit=143&less on=149)
- ☐ Snakes and Ladders - Not on the Board - Part 06 (unit? unit=143&less on=150)
- ☐ Spiral Traversing - Let's Animate (unit? unit=143&less on=151)

```
def spiral(row, column, arr) :
    rowStart = 0; columnStart = 0

    while (rowStart < row and columnStart < column) :

        for i in range(rowStart, row) :
            print(arr[i][columnStart], end = " ")

            columnStart = columnStart + 1

        for i in range (columnStart, column) :
            print( arr[row - 1][i], end = " ")

        row = row - 1

        if (rowStart < row) :
            for i in range(row - 1, rowStart - 1, -1) :
                print(arr[i][column - 1], end = " ")
                column = column - 1

            if (columnStart < column) :
                for i in range(column - 1, columnStart - 1, -1) :
                    print( arr[rowStart][i], end = " ")

                rowStart = rowStart + 1

matrix = [ [ 1, 2, 3 ],
            [ 5, 6, 7 ],
            [ 9, 10, 11 ] ]

row = 3
column = 3

spiral(row, column, matrix)
```

- ☐ 1, 2, 3, 7, 11, 10, 9, 5, 6
- ☐ 1, 2, 3, 5, 6, 7, 9, 10, 11
- ☒ 1, 5, 9, 10, 11, 7, 3, 2, 6
- ☐ 1, 5, 9, 2, 6, 10, 3, 7, 11

Yes, the answer is correct.

Score: 1

Accepted Answers:

1, 5, 9, 10, 11, 7, 3, 2, 6

3) What type of triangle is drawn by the given code?

1 point

```
import turtle

pen = turtle.Turtle()

for i in range(3):

    pen.forward(60)
    pen.right(120)

turtle.done()
```

- ☐ Scalar triangle
- ☐ Right angle triangle
- ☒ Equilateral triangle
- ☐ Isosceles triangle

☐ Spiral Traversing - Let's Animate - Part 01 (unit? unit=143&less on=152)

☐ Spiral Traversing - Let's Animate - Part 02 (unit? unit=143&less on=153)

☐ Spiral Traversing - Let's Animate - Part 03 (unit? unit=143&less on=154)

☐ Spiral Traversing - Let's Animate - Part 04 (unit? unit=143&less on=155)

☐ Spiral Traversing - Let's Animate - Part 05 (unit? unit=143&less on=156)

☐ Spiral Traversing - Let's Animate - Part 06 (unit? unit=143&less on=157)

☐ Spiral Traversing - Let's Animate - Part 07 (unit? unit=143&less on=158)

☐ GPS - Track the route (unit? unit=143&less on=159)

☐ GPS - Track the route - Part 01 (unit?

Yes, the answer is correct.

Score: 1

Accepted Answers:

Equilateral triangle

4) Which of the following programs will draw a hexagon?

1 point

☐

```
1 import turtle
2
3 pen = turtle.Turtle()
4
5 for i in range(3):
6
7     pen.right(60)
8     pen.forward(60)
9
10 turtle.done()
```

☐

```
1 import turtle
2
3 pen = turtle.Turtle()
4
5 for i in range(3):
6
7     pen.right(60)
8     pen.forward(60)
9     pen.right(60)
10    pen.forward(60)
11
12 turtle.done()
```

☐

```
1 import turtle
2
3 pen = turtle.Turtle()
4
5 for i in range(6):
6
7     pen.right(30)
8     pen.forward(60)
9
10 turtle.done()
```

☐

```
1 import turtle
2
3 pen = turtle.Turtle()
4
5 for i in range(6):
6
7     pen.right(30)
8     pen.forward(30)
9
10 turtle.done()
```

Yes, the answer is correct.

unit=143&less
on=160)

☐ GPS - Track
the route - Part
02 (unit?
unit=143&less
on=161)

☐ GPS - Track
the route - Part
03 (unit?
unit=143&less
on=162)

☐ GPS - Track
the route - Part
04 (unit?
unit=143&less
on=163)

☐ Week 7
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=143&less
on=164)

☒ **Quiz: Week 7
: Assignment
7
(assessment?
name=533)**

☐ Week 7:
Programming
Assignment 1
(/noc25_cs69/
progassignme
nt?name=527)

☐ Week 7:
Programming
Assignment 2
(/noc25_cs69/
progassignme
nt?name=528)

☐ Week 7:
Programming
Assignment 3
(/noc25_cs69/
progassignme
nt?name=529)

Week 8 ()

Score: 1

Accepted Answers:

```
1  import turtle
2
3  pen = turtle.Turtle()
4
5  for i in range(3):
6
7      pen.right(60)
8      pen.forward(60)
9      pen.right(60)
10     pen.forward(60)
11
12  turtle.done()
```

5) What will be the output of the following code?

1 point

```
my_dict = {}
my_dict[1] = "one"
my_dict[1.0] = "float one"
my_dict[True] = "boolean one"

print(my_dict)
```

- ☐ {1: "one", 1.0: "float one", True: "boolean one"}
- ☒ {1: "boolean one"}
- ☐ {1: "one", 1.0: "float one"}
- ☐ {1.0: "float one", True: "boolean one"}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{1: "boolean one"}

6) What is the output of the following code?

1 point

```
1  import turtle
2
3  pen = turtle.Turtle()
4
5  for i in range(3):
6
7      pen.right(60)
8      pen.forward(60)
9      pen.penup()
10     pen.right(60)
11     pen.forward(60)
12     pen.pendown()
13
14
15  turtle.done()
```

Week 9 ()

Week 10 ()

Week 11 ()

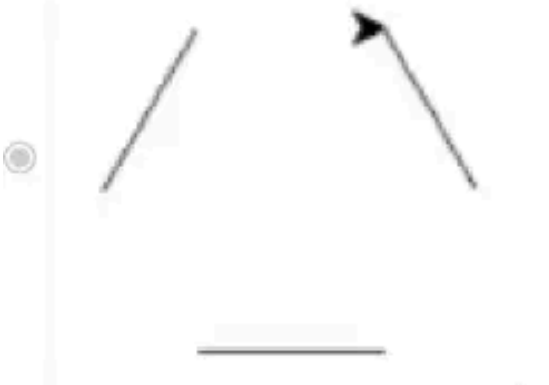
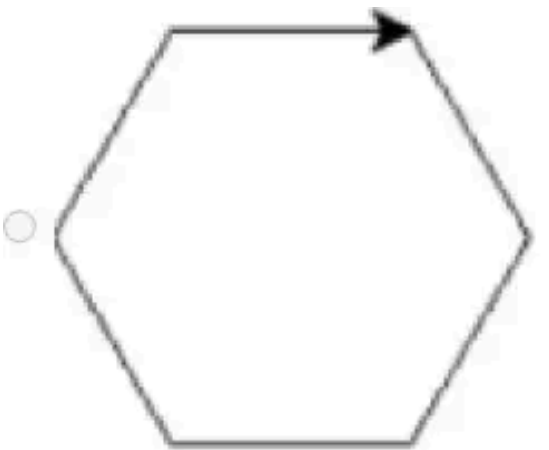
Week 12 ()

Text Transcripts ()

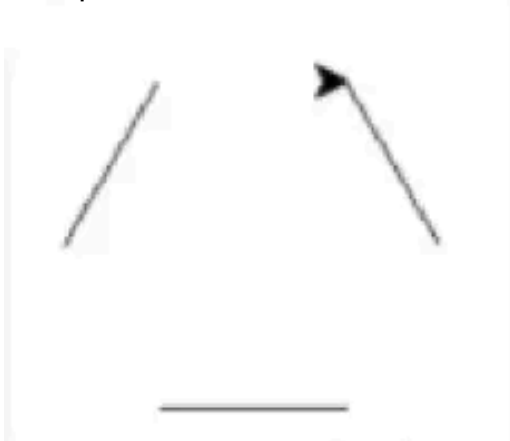
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Books ()

Problem Solving Session - Jan 2025 ()



Yes, the answer is correct.
Score: 1
Accepted Answers:



7) Which turtle command is equivalent to lifting up a pen?

1 point

- ☐ penlift()
- ☒ penup()
- ☐ uppen()
- ☐ penremove()

Yes, the answer is correct.

Score: 1

Accepted Answers:

penup()

8) Why do we use functions?

1 point

- ☐ To improve readability.
- ☐ To reuse code blocks.
- ☐ For the ease of code debugging.
- ☒ All of the above.

Yes, the answer is correct.

Score: 1

Accepted Answers:

All of the above.

9) Which library is used to import images?

1 point

- ☒ PIL
- ☐ Imageview
- ☐ IMG
- ☐ image

Yes, the answer is correct.

Score: 1

Accepted Answers:

PIL

10) In Snakes and Ladders, what can be the ways to track ladders and snakes?

0 points

- ☐ Maintain a dictionary with snake or ladder number blocks as keys.
- ☐ Using the if condition to check on every number.
- ☒ Both A and B.
- ☐ Both A and B.

Yes, the answer is correct.

Score: 0

Accepted Answers:

Both A and B.

X



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Course
outline

About NPTEL
()

How does an
NPTEL online
course work?
()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 8 : Assignment 8

The due date for submitting this assignment has passed.

Due on 2025-03-19, 23:59 IST.

Assignment submitted on 2025-03-19, 21:34 IST

1) What will be the output of the following Python code?

1 point

```
tup = ([1, 2], [3, 4])
tup[0].append(5)
print(tup)
```

- ☒ ([1, 2, 5], [3, 4])
- ☐ ([1, 2], [3, 4])
- ☐ ([1, 2], [3, 4, 5])
- ☐ TypeError: Tuple object does not support item assignment

Yes, the answer is correct.
Score: 1
Accepted Answers:
([1, 2, 5], [3, 4])

2) What operations can be performed on tuples?

1 point

- ☐ Tuples are appendable
- ☐ We can delete a value from tuples
- ☐ Both (a) and (b)
- ☒ We can count the number of instances of an element

Yes, the answer is correct.
Score: 1
Accepted Answers:
We can count the number of instances of an element

☐ Tuples- Python Data Structure (unit? unit=165&lesson=166)

☐ Lottery Simulation - Profit or Loss (unit? unit=165&lesson=167)

☐ Lottery Simulation - Profit or Loss - Part 01 (unit? unit=165&lesson=168)

☐ Lottery Simulation - Profit or Loss - Part 02 (unit? unit=165&lesson=169)

☐ Lottery Simulation - Profit or Loss - Part 03 (unit? unit=165&lesson=170)

☐ Lottery Simulation - Profit or Loss - Part 04 (unit? unit=165&lesson=171)

☐ Lottery Simulation - Profit or Loss - Part 05 (unit? unit=165&lesson=172)

☐ Lottery Simulation - Profit or Loss - Part 06 (unit? unit=165&lesson=173)

☐ Image Processing - Enhance your images (unit? unit=165&lesson=174)

☐ Image Processing -

3) What will be the output of the following Python code?

1 point

```
t = (1,2,3,4,5)

for i in range(-1,-len(t), -1):
    print(t[i])
```

- ☐ 1,2,3,4,5
- ☐ 5,4,3,2,1
- ☒ 5,4,3,2
- ☐ 1,2,3,4

Yes, the answer is correct.

Score: 1

Accepted Answers:

5,4,3,2

4) What will be the output of the following Python code?

1 point

```
1 word = 'facebook'
2 new = ''
3
4 for w in word:
5     i = ord(w)
6
7     j = ((( i+26 ) - 97 ) % 26 ) + 97
8
9     new = new + chr(j)
10
11 print(new)
```

- ☒ facebook
- ☐ gbdfcppl
- ☐ ezbdannj
- ☐ ytvxuhhd

Yes, the answer is correct.

Score: 1

Accepted Answers:

facebook

5) When will the program print "Clap!"?

1 point

Enhance your images - Part 01 (unit? unit=165&lesson=175)

☐ Image Processing - Enhance your images - Part 02 (unit? unit=165&lesson=176)

☐ Image Processing - Enhance your images - Part 03 (unit? unit=165&lesson=177)

☐ Anagrams (unit? unit=165&lesson=178)

☐ Anagrams - Part 01 (unit? unit=165&lesson=179)

☐ Anagrams - Part 02 (unit? unit=165&lesson=180)

☐ Anagrams - Part 03 (unit? unit=165&lesson=181)

☐ Facebook Sentiment Analysis (unit? unit=165&lesson=182)

☐ Facebook Sentiment Analysis - Part 01 (unit? unit=165&lesson=183)

☐ Facebook Sentiment Analysis - Part 02 (unit? unit=165&lesson=184)

☐ Facebook Sentiment Analysis - Part 03 (unit?

```

1  print('Enter a letter between a-z')
2
3  player1 = str(input())
4  player2 = str(input())
5
6  while(True):
7
8      if(ord(player1.lower())+1 == ord(player2.lower())):
9          print('clap')
10         break
11     else:
12         player1 = str(input())
13         player2 = str(input())
14

```

- ☐ When both players enter the same letters.
- ☒ When player 2 enters the next letter after player 1.
- ☐ When player 1 enters the next letter after player 2.
- ☐ It will never clap.

Yes, the answer is correct.

Score: 1

Accepted Answers:

When player 2 enters the next letter after player 1.

6) What will be the effect of the following Python code on the graph?

1 point

```

1  import random
2  import matplotlib.pyplot as plt
3
4  l = []
5  count = 0
6
7  for i in range(10):
8      guess = random.randint(1, 10)
9      pick = random.randint(1, 10)
10
11     if(guess!=pick):
12         count+=1
13         l.append(count)
14     else:
15         count-=1
16         l.append(count)
17
18  plt.plot(l)
19  plt.show()

```

- ☐ The graph will go up when guess and pick are the same.
- ☐ The graph will go down when guess and pick are the same.

unit=165&lesson=185)

☐ Facebook Sentiment Analysis - Part 04 (unit? unit=165&lesson=186)

☒ Week 8: Programming Assignment 1 (/noc25_cs69/progassignment?name=530)

☐ Week 8: Programming Assignment 2 (/noc25_cs69/progassignment?name=531)

☒ Week 8: Programming Assignment 3 (/noc25_cs69/progassignment?name=532)

☐ Week 8 Feedback Form: The Joy of Computing using Python (unit? unit=165&lesson=187)

☒ Quiz: Week 8 : Assignment 8 (assessment? name=535)

Week 9 ()

Week 10 ()

Week 11 ()

Week 12 ()

Text Transcripts ()

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Books ()

- ☐ The graph will go up when guess and pick are not the same.
- ☒ Both (b) and (c).

Yes, the answer is correct.
Score: 1

Accepted Answers:
Both (b) and (c).

7) What is the primary advantage of using tokenization in Natural Language Processing (NLP)? **1 point**

- ☐ It allows the model to understand entire sentences as a single entity.
- ☒ It splits text into smaller, manageable units, which helps in processing language more effectively.
- ☐ It eliminates the need for syntactical analysis.
- ☐ It improves the grammatical structure of the text.

Yes, the answer is correct.
Score: 1

Accepted Answers:

It splits text into smaller, manageable units, which helps in processing language more effectively.

8) What will be the output of the following Python code? **1 point**

```
1 string = 'hey!there'
2
3 a = sorted(string)
4 a.reverse()
5
6 print(a)
```

- ☐ ['!', 'e', 'e', 'e', 'h', 'h', 'r', 't', 'y']
- ☐ ['h', 'e', 'y', '!', 't', 'h', 'e', 'r', 'e']
- ☒ ['y', 't', 'r', 'h', 'h', 'e', 'e', 'e', '!']
- ☐ None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:

['y', 't', 'r', 'h', 'h', 'e', 'e', 'e', '!']

9) While converting an image into black and white, can it be converted back into a colored image? **1 point**

- ☐ True
- ☒ False

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

10) What will be the output of the following Python code? **1 point**

**Problem
Solving
Session - Jan
2025 ()**

```
def test(word):  
    new_word = ''  
    for ch in word:  
        if ch>='a' and ch<='z':  
            temp = ord(ch)  
            temp = temp - 32  
            temp = chr(temp)  
            new_word = new_word + temp  
    return new_word
```

- ☒ Converting lowercase letters into uppercase.
- ☐ Converting uppercase letters into lowercase.
- ☐ Returns the same word.
- ☐ Error.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Converting lowercase letters into uppercase.

X



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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 9 : Assignment 9

The due date for submitting this assignment has passed.

Due on 2025-03-26, 23:59 IST.

Assignment submitted on 2025-03-26, 21:25 IST

1) How can we identify which book is written by which author? 1 point

- ☐ By matching handwriting.
- ☒ By analyzing word length with previous books.
- ☐ By analyzing the number of pages in a book.
- ☐ By analyzing the book's preface.

Yes, the answer is correct.
Score: 1

Accepted Answers:
By analyzing word length with previous books.

2) What is the key difference between lists and tuples in Python, and when would you choose one over the other? 1 point

- ☐ Lists are immutable and can be used when elements need to be modified, while tuples are mutable and best for storing fixed data.
- ☒ Lists are mutable and can be used for dynamic data manipulation, while tuples are immutable and used to store fixed collections of items.
- ☐ Both lists and tuples are mutable and can be used interchangeably for any purpose.
- ☐ Tuples are used only for string data, while lists can store any data type.

Yes, the answer is correct.
Score: 1

Accepted Answers:
Lists are mutable and can be used for dynamic data manipulation, while tuples are immutable and used to store fixed collections of items.

Week 7 ()

Week 8 ()

Week 9 ()

- ☐ Natural Language Processing - Author Stylometry (unit? unit=188&less on=189)
- ☐ Natural Language Processing - Author Stylometry - Part 01 (unit? unit=188&less on=190)
- ☐ Natural Language Processing - Author Stylometry - Part 02 (unit? unit=188&less on=191)
- ☐ Natural Language Processing - Author Stylometry - Part 03 (unit? unit=188&less on=192)
- ☐ Natural Language Processing - Author Stylometry - Part 04 (unit? unit=188&less on=193)
- ☐ Natural Language Processing - Author Stylometry - Part 05 (unit?

3) Will the following piece of code always return True?

1 point

```
import networkx as nx
G = nx.gnp_random_graph(10, 0.5)
print(nx.is_connected(G))
```

- ☐ True
☒ False

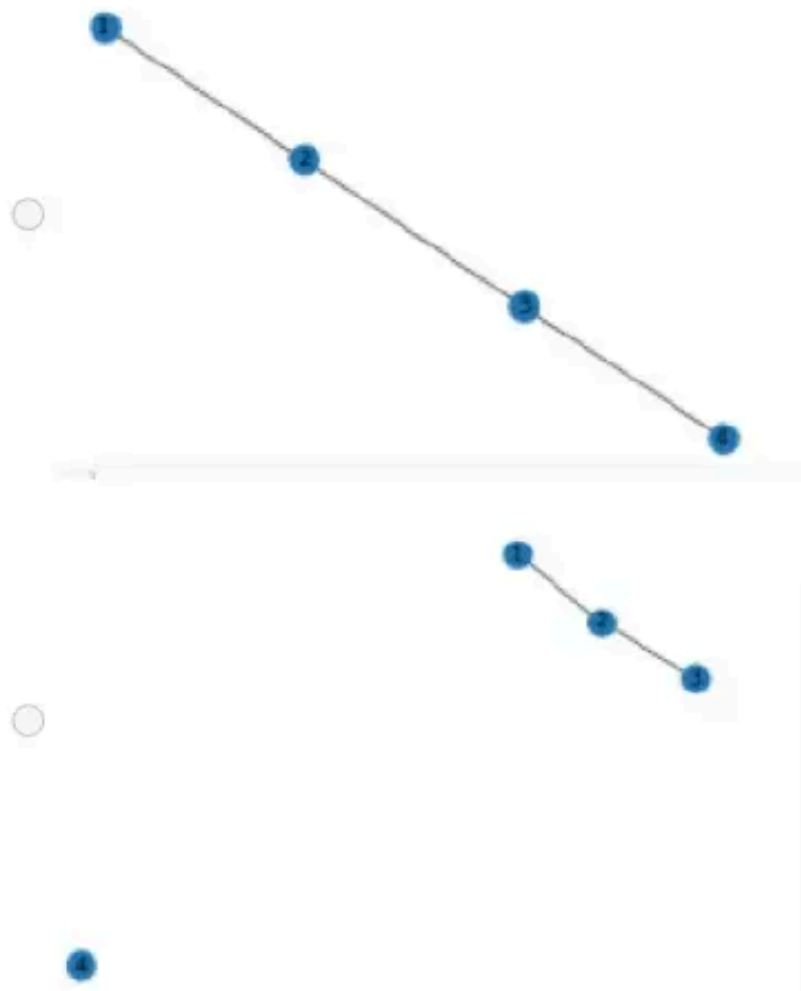
Yes, the answer is correct.
Score: 1

Accepted Answers:
False

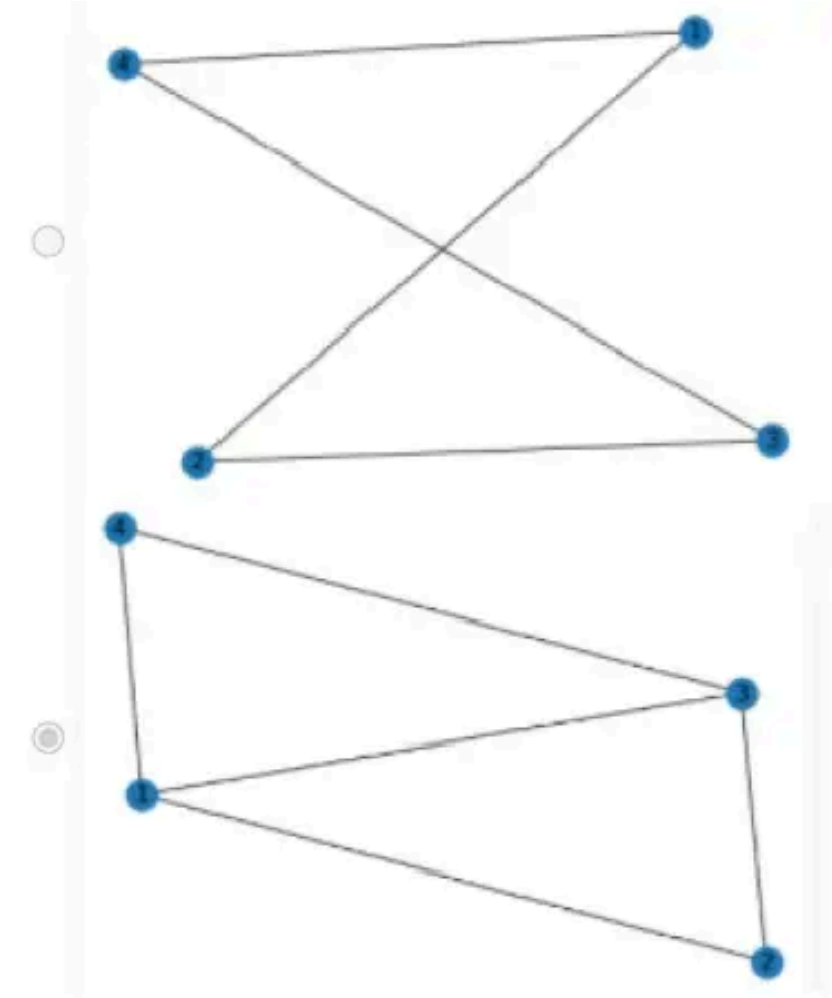
4) What is the output of the following code?

1 point

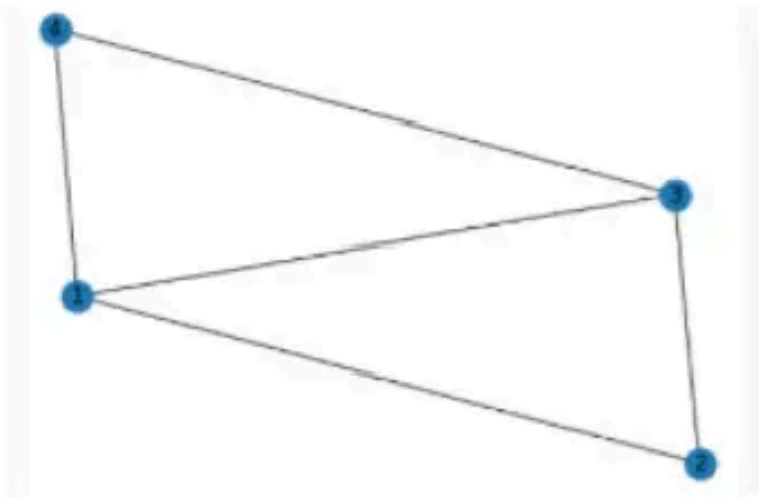
```
1 import networkx as nx
2 import matplotlib.pyplot as plt
3
4
5 G = nx.Graph()
6 G.add_nodes_from([1, 2, 3, 4])
7 G.add_edges_from([(1, 2), (2, 1), (2, 3), (3, 4), (4, 1), (3, 1)])
8
9 nx.draw(G, with_labels=True)
10 plt.show()
```



- ☐ unit=188&less on=194)
- ☐ Natural Language Processing - Author Stylometry - Part 06 (unit? unit=188&less on=195)
- ☐ Natural Language Processing - Author Stylometry - Part 07 (unit? unit=188&less on=196)
- ☐ Natural Language Processing - Author Stylometry - Part 08 (unit? unit=188&less on=197)
- ☐ Natural Language Processing - Author Stylometry - Part 09 (unit? unit=188&less on=198)
- ☐ Natural Language Processing - Author Stylometry - Part 10 (unit? unit=188&less on=199)
- ☐ Introduction to Networkx - Part 01 (unit? unit=188&less on=200)
- ☐ Introduction to Networkx - Part 02 (unit? unit=188&less on=201)

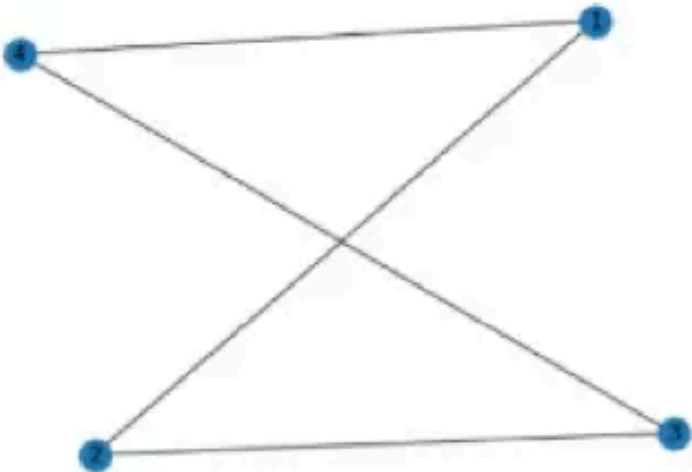


Yes, the answer is correct.
Score: 1
Accepted Answers:



5) What is the total degree of the given graph?

1 point



- ☐ 10
- ☒ 8

☐ Six Degrees of Separation : Meet your favourites (unit? unit=188&less on=202)

☐ Six Degrees of Separation : Meet your favourites - Part 01 (unit? unit=188&less on=203)

☐ Six Degrees of Separation : Meet your favourites - Part 02 (unit? unit=188&less on=204)

☐ Six Degrees of Separation : Meet your favourites - Part 03 (unit? unit=188&less on=205)

☐ Area Calculation - Don't Measure (unit? unit=188&less on=206)

☐ Area Calculation - Don't Measure - Part 01 (unit? unit=188&less on=207)

☐ Area Calculation - Don't Measure - Part 02 (unit? unit=188&less on=208)

☐ Area Calculation - Don't Measure - Part 03 (unit? unit=188&less on=209)

☐ 6
☐ 4

Yes, the answer is correct.

Score: 1

Accepted Answers:
8

6) Which of the following is always true for a connected undirected graph with at least one edge? **1 point**

- ☐ Every node has an even degree
☒ The sum of all node degrees is even
☐ The graph contains an Eulerian path
☐ The graph contains a Hamiltonian cycle

Yes, the answer is correct.

Score: 1

Accepted Answers:
The sum of all node degrees is even

7) What is the primary difference between using single quotes, double quotes, and triple quotes to create strings in Python? **1 point**

- ☒ Single and double quotes can only be used for single-line strings, while triple quotes are used for multi-line strings
☐ Single quotes are used for strings with one word, while double quotes are for strings with multiple words
☐ Double quotes are more efficient than single quotes for creating strings
☐ There is no difference in how Python interprets single quotes, double quotes, or triple quotes for strings

Yes, the answer is correct.

Score: 1

Accepted Answers:
Single and double quotes can only be used for single-line strings, while triple quotes are used for multi-line strings

8) How many nodes and edges does the following graph have? **1 point**



☒ 10,6
☐ 10,5

☐ Area Calculation - Don't Measure - Part 04 (unit? unit=188&lesson=210)

☐ Area Calculation - Don't Measure - Part 05 (unit? unit=188&lesson=211)

☐ Area Calculation - Don't Measure - Part 06 (unit? unit=188&lesson=212)

☐ Week 9 Feedback Form: The Joy of Computing using Python (unit? unit=188&lesson=213)

☒ Week 9: Programming Assignment 1 (/noc25_cs69/progassignment?name=534)

☒ Week 9: Programming Assignment 2 (/noc25_cs69/progassignment?name=536)

☒ Week 9: Programming Assignment 3 (/noc25_cs69/progassignment?name=537)

☒ **Quiz: Week 9 : Assignment 9 (assessment? name=538)**

Week 10 ()

- ☐ 8,6
☐ 8,5

Yes, the answer is correct.

Score: 1

Accepted Answers:

10,6

9) A complete graph will have a degree of separation?

1 point

- ☒ 1
☐ 2
☐ 3
☐ Depends on the number of nodes

Yes, the answer is correct.

Score: 1

Accepted Answers:

1

10) What will be the output of the following code?

1 point

```
s = "hello"
s[0] = "H"
```

- ☐ It will print "Hello"
☒ It will raise a TypeError
☐ The string will be updated to "Hello"
☐ It will change the first character to uppercase and print "Hello"

Yes, the answer is correct.

Score: 1

Accepted Answers:

It will raise a TypeError

Week 11 ()
Week 12 ()
Text Transcripts ()
Download Videos ()
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Problem Solving Session - Jan 2025 ()

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Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 10 : Assignment 10

The due date for submitting this assignment has passed.

Due on 2025-04-02, 23:59 IST.

Assignment submitted on 2025-04-02, 22:25 IST

1) Which of the following statements is correct about Python's dynamic typing? **1 point**

- ☐ Variables must be declared before use.
- ☒ A variable's type can change at runtime.
- ☐ Python does not allow reassignment of variable types.
- ☐ Python is statically typed.

Yes, the answer is correct.

Score: 1

Accepted Answers:

A variable's type can change at runtime.

2) What will be the output of the following code? **1 point**

```
s = "PythonProgramming"
print(s[2:15:3] + s[::-4])
```

- ☐ toPmiagmnrh
- ☐ toPmiagmnrPg
- ☒ tnoaigmooP
- ☐ toPmiagmnr

Yes, the answer is correct.

Week 7 ()**Week 8 ()****Week 9 ()****Week 10 ()**

☐ FLAMES -
Part 01 (unit?
unit=214&less
on=215)

☐ FLAMES -
Part 02 (unit?
unit=214&less
on=216)

☐ FLAMES -
Part 03 (unit?
unit=214&less
on=217)

☐ FLAMES -
Part 04 (unit?
unit=214&less
on=218)

☐ FLAMES -
Part 05 (unit?
unit=214&less
on=219)

☐ FLAMES -
Part 06 (unit?
unit=214&less
on=220)

☐ Data
Compression -
Part 01 (unit?
unit=214&less
on=221)

☐ Data
Compression -
Part 02 (unit?
unit=214&less
on=222)

☐ Data
Compression -
Part 03 (unit?
unit=214&less
on=223)

☐ Data
Compression -
Part 04 (unit?

Score: 1

Accepted Answers:

tnoaignmooP

3) What does the following code print?

1 point

```
s = "Python Programming"
print(s.index('o', 5, 10))
```

- ☐ 6
- ☐ 7
- ☒ 9
- ☐ Value Error

Yes, the answer is correct.

Score: 1

Accepted Answers:

9

4) Which statement about find() and index() in Python is true?

1 point

- ☐ Both return -1 if the substring is not found.
- ☒ index() raises an error if not found, but find() returns -1.
- ☐ index() and find() behave identically.
- ☐ Both return the first index of occurrence.

Yes, the answer is correct.

Score: 1

Accepted Answers:

index() raises an error if not found, but find() returns -1.

5) What will be the output of the following Python code?

1 point

```
print(17 % -5)
```

- ☐ 3
- ☐ 2
- ☐ 0
- ☒ -3

Yes, the answer is correct.

Score: 1

Accepted Answers:

-3

6) Which method in PIL is used to resize an image while maintaining the aspect ratio? **1 point**

- ☐ resize()
- ☒ thumbnail()
- ☐ reshape()

unit=214&less
on=224)

- ☐ Data
Compression -
Part 05 (unit?
unit=214&less
on=225)

**Quiz: Week
10 :
Assignment
10
(assessment?
name=549)**

- Week 10 :**
Programming
Assignment 1
(/noc25_cs69/
progassignme
nt?name=539)

- Week 10:**
Programming
Assignment 2
(/noc25_cs69/
progassignme
nt?name=540)

- Week 10:**
Programming
Assignment 3
(/noc25_cs69/
progassignme
nt?name=541)

- Week 10**
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=214&less
on=226)

Week 11 ()

Week 12 ()

**Text
Transcripts
()**

**Download
Videos ()**

- ☐ aspect ratio resize()

Yes, the answer is correct.

Score: 1

Accepted Answers:
thumbnail()

7) Which of the following correctly finds all numbers divisible by 3 but not by 5 in a list? **1 point**

nums = range(1, 50)

- ☒ [x for x in nums if x % 3 == 0 and x % 5 != 0]
☐ [x for x in nums if x % 5 != 0 or x % 3 == 0]
☐ [x for x in nums if x % 3 != 0 and x % 5 == 0]
☐ filter(lambda x: x % 3 == 0 and x % 5 == 0, nums)

Yes, the answer is correct.

Score: 1

Accepted Answers:

[x for x in nums if x % 3 == 0 and x % 5 != 0]

8) What will be the output of the following code?

1 point

```
import numpy as np
arr = np.array([[1, [2, 3]], [4, [5, 6]], [7, [8, 9]]], dtype=object)
print(arr[1][1][0])
```

- ☐ 2
☒ 5
☐ [5, 6]
☐ 4

Yes, the answer is correct.

Score: 1

Accepted Answers:

5

9) For which value of axis the output is [5, 7, 9] in the below code?

1 point

```
import numpy as np
a = np.array([[1, 2, 3], [4, 5, 6]])
print(np.sum(a, axis= ))
```

- ☒ 0
☐ 1
☐ 2
☐ axis argument is not required.

Yes, the answer is correct.

Books ()**Problem
Solving
Session -
Jan 2025 ()**

Score: 1

Accepted Answers:

0

10) What will be the shape of the resulting NumPy array?

1 point

```
import numpy as np
a = np.ones((3, 4))
b = np.ones((4, 2))
c = np.dot(a, b)
print(c.shape)
```

- ☒ (3, 2)
- ☐ (4, 3)
- ☐ (3, 4)
- ☐ (4, 2)

Yes, the answer is correct.

Score: 1

Accepted Answers:

(3, 2)

X



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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 11 : Assignment

The due date for submitting this assignment has passed.

Due on 2025-04-09, 23:59 IST.

Assignment submitted on 2025-04-09, 15:26 IST

1) In Selenium, what is the purpose of WebDriver, such as webdriver.Chrome()? **1 point**

- ☐ It is used to interact with web elements like buttons and text fields.
- ☐ It launches and controls a web browser session for automation.
- ☒ It is used to write test cases in Selenium.
- ☐ It provides built-in methods to handle databases in web applications.

No, the answer is incorrect.

Score: 0

Accepted Answers:

It launches and controls a web browser session for automation.

2) Which of the following is the best locator strategy in Selenium in terms of performance? **1 point**

- ☐ find_element_by_name()
- ☒ find_element_by_id()
- ☐ find_element_by_class_name()
- ☐ find_element_by_xpath()

Yes, the answer is correct.

Score: 1

Accepted Answers:

find_element_by_id()

3) Selenium WebDriver can be used to automate both web and native mobile applications on Android and iOS. **1 point**

Week 7 ()**Week 8 ()****Week 9 ()****Week 10 ()****Week 11 ()**

☐ Browser Automation Whatsapp using Python - Part 01 (unit? unit=227&less on=228)

☐ Browser Automation Whatsapp using Python - Part 02 (unit? unit=227&less on=229)

☐ Browser Automation Whatsapp using Python - Part 03 (unit? unit=227&less on=230)

☐ Browser Automation Whatsapp using Python - Part 04 (unit? unit=227&less on=231)

☐ Fun with Calendar - Part 01 (unit? unit=227&less on=232)

☐ Fun with Calendar - Part 02 (unit? unit=227&less on=233)

☐ Fun with Calendar - Part 03 (unit?

- ☐ True
☒ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

False

4) What is the output of the following Python code?

1 point

```
for i in range(5):
    for j in range(5):
        if j == 2:
            break
        if i == 3:
            continue
        print(f"({i}, {j})", end=" ")
```

- ☒ (0, 0) (0, 1) (1, 0) (1, 1) (2, 0) (2, 1) (4, 0) (4, 1)
☐ (0, 0) (0, 1) (1, 0) (1, 1) (2, 0) (2, 1) (3, 0) (3, 1) (4, 0) (4, 1)
☐ (0, 0) (0, 1) (1, 0) (1, 1) (2, 0) (2, 1) (3, 0) (3, 1) (3, 2) (4, 0) (4, 1)
☐ (0, 0) (0, 1) (1, 0) (1, 1) (2, 0) (2, 1) (3, 0) (4, 0) (4, 1) (4, 2)

Yes, the answer is correct.

Score: 1

Accepted Answers:

(0, 0) (0, 1) (1, 0) (1, 1) (2, 0) (2, 1) (4, 0) (4, 1)

5) What will be the output of the following Python code, which performs a modified linear search?

1 point

unit=227&less
on=234)

☐ Fun with
Calendar -
Part 04 (unit?
unit=227&less
on=235)

☐ Fun with
Calendar -
Part 05 (unit?
unit=227&less
on=236)

☐ Fun with
Calendar -
Part 06 (unit?
unit=227&less
on=237)

☐ Fun with
Calendar -
Part 07 (unit?
unit=227&less
on=238)

☐ Fun with
Calendar -
Part 08 (unit?
unit=227&less
on=239)

☐ Fun with
Calendar -
Part 09 (unit?
unit=227&less
on=240)

☐ Fun with
Calendar -
Part 10 (unit?
unit=227&less
on=241)

☐ Fun with
Calendar -
Part 11 (unit?
unit=227&less
on=242)

☐ Fun with
Calendar -
Part 12 (unit?
unit=227&less
on=243)

☒ Week 11 :
Programming
Assignment 1

```
arr = [3, 8, 1, 7, 9, 2]
key = 7
```

```
for i in range(len(arr)):
    if arr[i] == key:
        print(f"Found at index {i}")
        break
    if arr[i] % 2 == 0:
        continue
    print(f"Checking {arr[i]}")
```

☐ Checking 3
Checking 8
Checking 1
Found at index 3

☒ Checking 3
Checking 1
Found at index 3

☐ Checking 3
Checking 8
Checking 1
Checking 7
Found at index 3

☐ Found at index 3

Yes, the answer is correct.
Score: 1

Accepted Answers:

Checking 3

Checking 1

Found at index 3

6) What does calendar.isleap(2100) return?

1 point

☒ False
☐ True
☐ None
☐ 1

Yes, the answer is correct.
Score: 1

Accepted Answers:

False

(/noc25_cs69/
progassignment?
name=542)

• Week 11 :
Programming
Assignment 2
(/noc25_cs69/
progassignment?
name=543)

• Week 11 :
Programming
Assignment 3
(/noc25_cs69/
progassignment?
name=544)

• Quiz: Week
11 :
Assignment
(assessment?
name=550)

• Week 11
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=227&less
on=244)

Week 12 ()

Text
Transcripts
()

Download
Videos ()

Books ()

Problem
Solving
Session -
Jan 2025 ()

7) What is the output of `datetime.date(2024, 2, 29) + datetime.timedelta(days=1)`? **1 point**

- ☐ 2024-02-30
☒ 2024-03-01
☐ ValueError
☐ 2024-02-29

Yes, the answer is correct.

Score: 1

Accepted Answers:

2024-03-01

8) Which function is used to get the current local time in Python? **1 point**

- ☒ `time.localtime()`
☐ `datetime.now()`
☐ `time.time()`
☐ `datetime.utcnow()`

Partially Correct.

Score: 0.5

Accepted Answers:

`time.localtime()`

`datetime.now()`

9) What happens when you apply `.astimezone()` to a naive datetime object? **1 point**

- ☐ It changes to UTC
☐ It raises an error
☒ It assumes local timezone
☐ It converts to IST

Yes, the answer is correct.

Score: 1

Accepted Answers:

It assumes local timezone

10) What is the output of the following Python code? **1 point**

```
from datetime import datetime
import pytz

dt_naive = datetime(2024, 2, 29, 12, 0, 0)
tz_ist = pytz.timezone('Asia/Kolkata')

dt_aware = tz_ist.localize(dt_naive)
print(dt_aware.strftime("%Y-%m-%d %H:%M:%S %Z"))
```

- ☒ 2024-02-29 12:00:00 IST

- ☐ 2024-02-29 06:30:00 UTC
- ☐ 2024-02-29 12:00:00 UTC
- ☐ Error

Yes, the answer is correct.

Score: 1

Accepted Answers:

2024-02-29 12:00:00 IST

X

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

week 4 ()

Week 5 ()

Week 6 ()

Week 12 : Assignment 12

The due date for submitting this assignment has passed.

Due on 2025-04-16, 23:59 IST.

As per our records you have not submitted this assignment.

1) Which of the following graphs is generated by the given Python code using the networkx module?

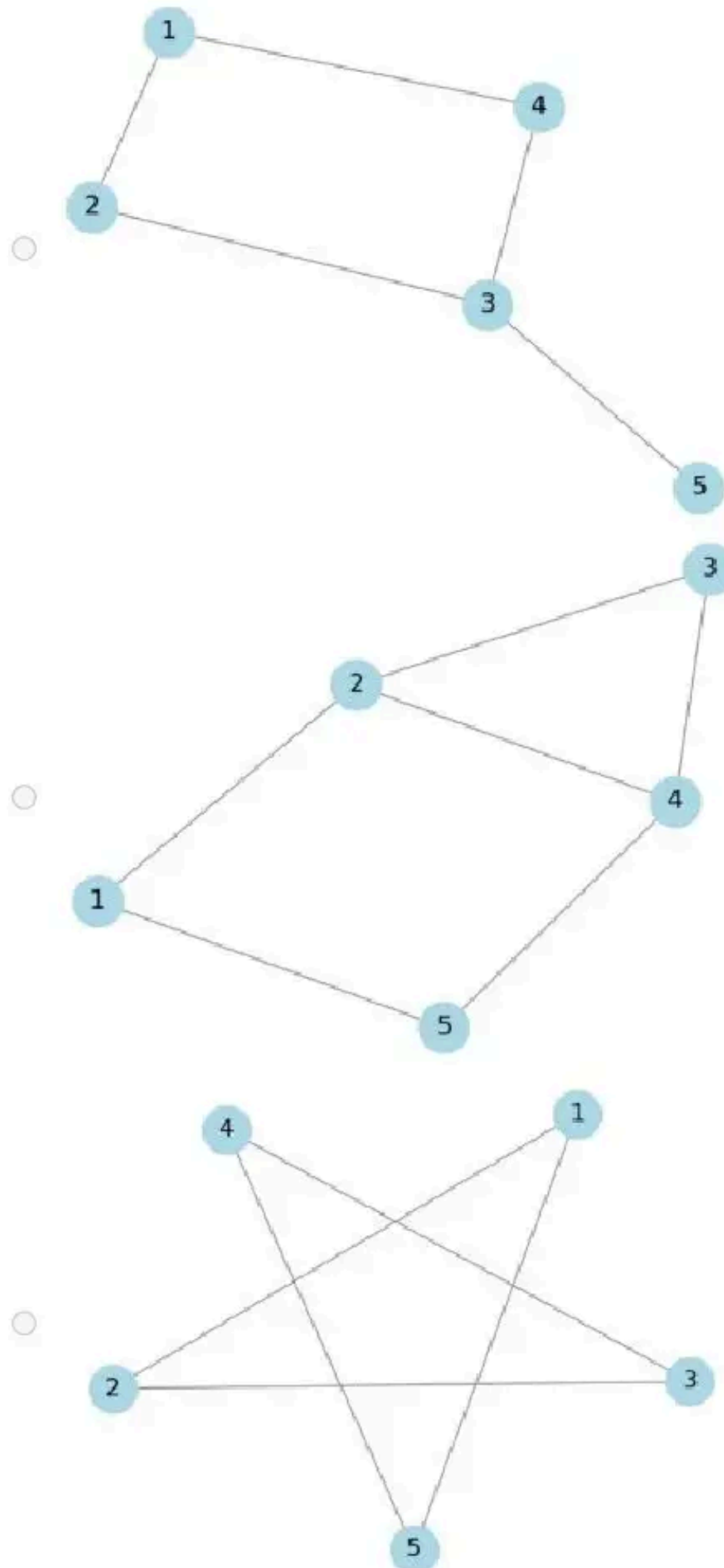
1 point

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.Graph()
G.add_nodes_from([1, 2, 3, 4, 5])
G.add_edges_from([(1, 2), (2, 3), (3, 4), (4, 5), (5, 1)])
nx.draw(G, with_labels=True, node_color='lightblue',
edge_color='gray', node_size=1000, font_size=16)
plt.show()
```

Week 7 ()**Week 8 ()****Week 9 ()****Week 10 ()****Week 11 ()****Week 12 ()**

- ☒ Page Rank -
How does
Google Work ?
- Part 01 (unit?
unit=245&less
on=246)
- ☐ Page Rank -
How does
Google Work ?
- Part 02 (unit?
unit=245&less
on=247)
- ☐ Page Rank -
How does
Google Work ?
- Part 03 (unit?
unit=245&less
on=248)
- ☐ Page Rank -
How does
Google Work ?
- Part 04 (unit?
unit=245&less
on=249)
- ☐ Page Rank -
How does
Google Work ?
- Part 05 (unit?
unit=245&less
on=250)
- ☐ Page Rank -
How does
Google Work ?
- Part 06 (unit?
unit=245&less
on=251)
- ☐ Page Rank -
How does
Google Work ?



- Part 07 (unit? unit=245&less on=252)

☐ Page Rank - How does Google Work ? - Part 08 (unit? unit=245&less on=253)

☐ Page Rank - How does Google Work ? - Part 09 (unit? unit=245&less on=254)

☐ Page Rank - How does Google Work ? - Part 10 (unit? unit=245&less on=255)

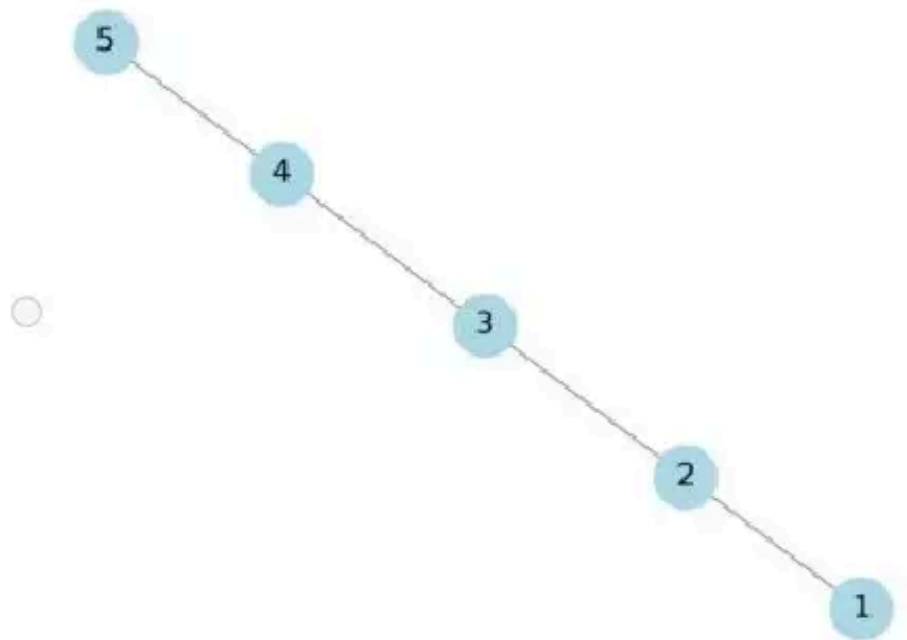
☐ Page Rank - How does Google Work ? - Part 11 (unit? unit=245&less on=256)

☐ Page Rank - How does Google Work ? - Part 12 (unit? unit=245&less on=257)

☐ Page Rank - How does Google Work ? - Part 13 (unit? unit=245&less on=258)

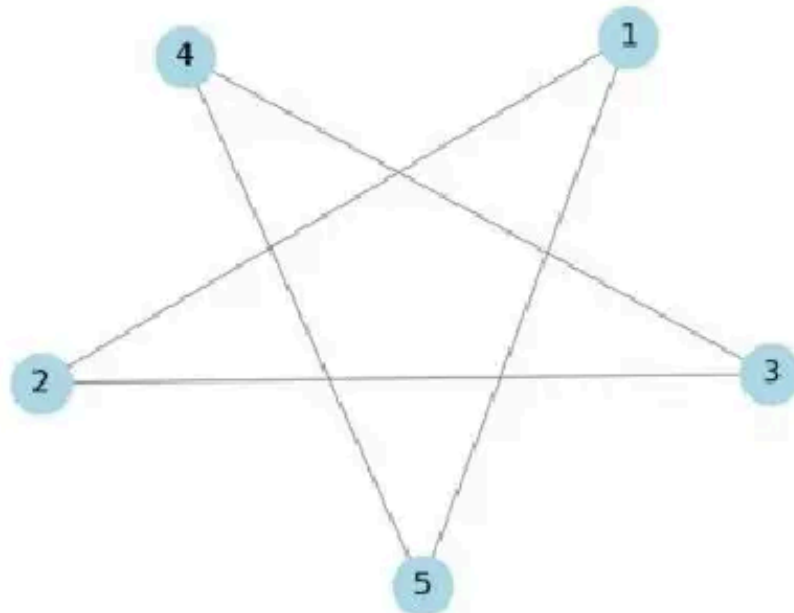
☐ Page Rank - How does Google Work ? - Part 14 (unit? unit=245&less on=259)

☐ Page Rank - How does Google Work ? - Part 15 (unit? unit=245&less on=260)



No, the answer is incorrect.
Score: 0

Accepted Answers:



2) In networkX, which function is used to return the degree of a specific node?

1 point

- ☐ G.degree(node)
☐ G.get_degree(node)
☐ G.node_degree(node)
☐ G.find_degree(node)

No, the answer is incorrect.
Score: 0

Accepted Answers:

G.degree(node)

3) Observe the following graph generated using the networkx module. How many edges does this graph contain?

1 point

○ Page Rank -
How does
Google Work ?
- Part 16 (unit?
unit=245&less
on=261)

○ Collatz
Conjecture -
Part 01 (unit?
unit=245&less
on=262)

○ Collatz
Conjecture -
Part 02 (unit?
unit=245&less
on=263)

○ JOC
Conclusion
(unit?
unit=245&less
on=264)

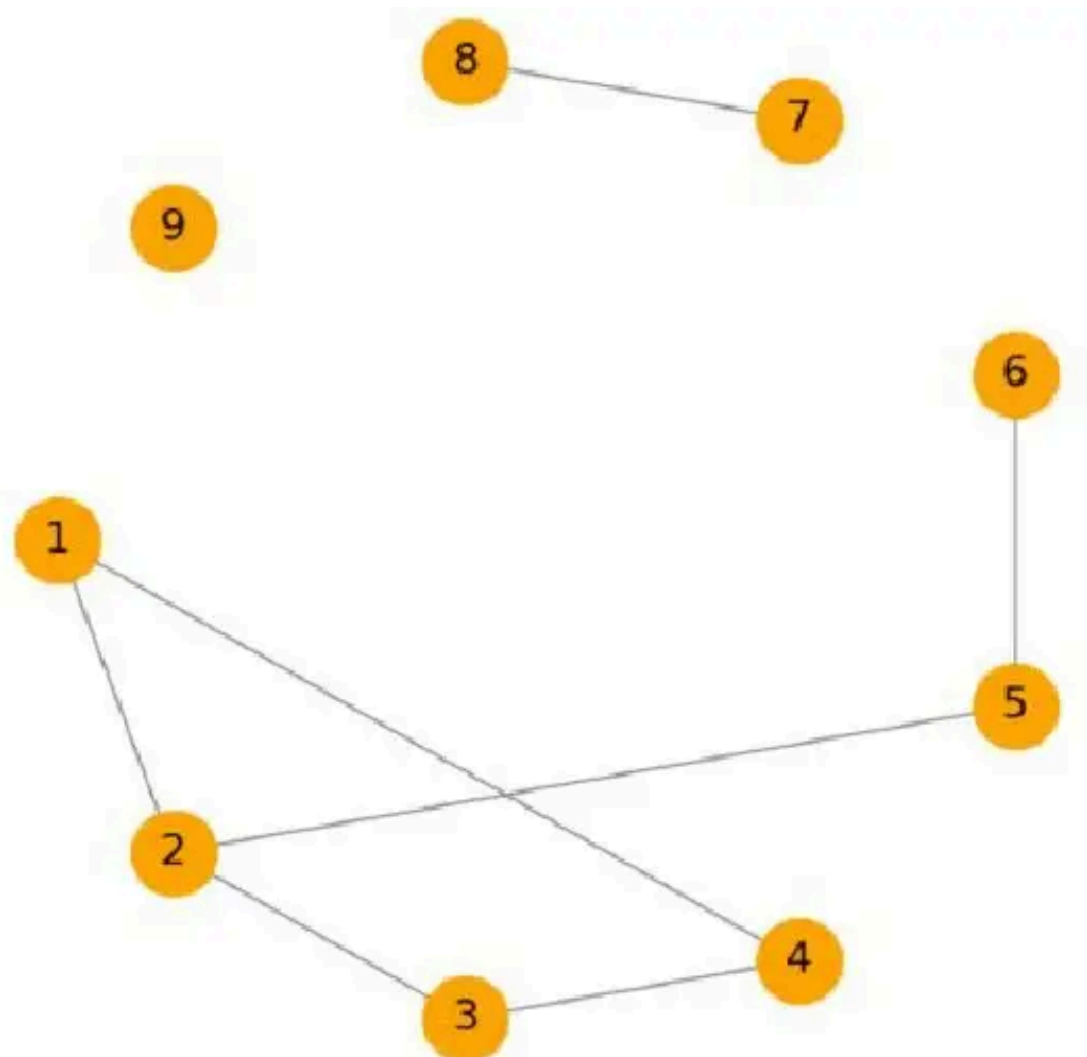
● Week 12 :
Programming
Assignment 1
(/noc25_cs69/
progassignme
nt?name=545)

● Week 12 :
Programming
Assignment 2
(/noc25_cs69/
progassignme
nt?name=546)

● Week 12 :
Programming
Assignment 3
(/noc25_cs69/
progassignme
nt?name=548)

○ Week 12
Feedback
Form: The Joy
of Computing
using Python
(unit?
unit=245&less
on=265)

○ Quiz: Week
12 :
Assignment
12



☐ 6

☐ 7

☐ 8

☐ 9

No, the answer is incorrect.

Score: 0

Accepted Answers:

7

4) What does `nx.cycle graph(4)` generate?

1 point

☐ A directed cycle of 4 nodes

☐ A cycle graph with 4 nodes connected in a closed loop

☐ A ladder graph with 4 rounds in the ladder

☐ A complete graph with 4 nodes

No, the answer is incorrect.

Score: 0

Accepted Answers:

A cycle graph with 4 nodes connected in a closed loop

5) Observe the following graph generated using the **networkx** module. What type of graph is this? 1 point

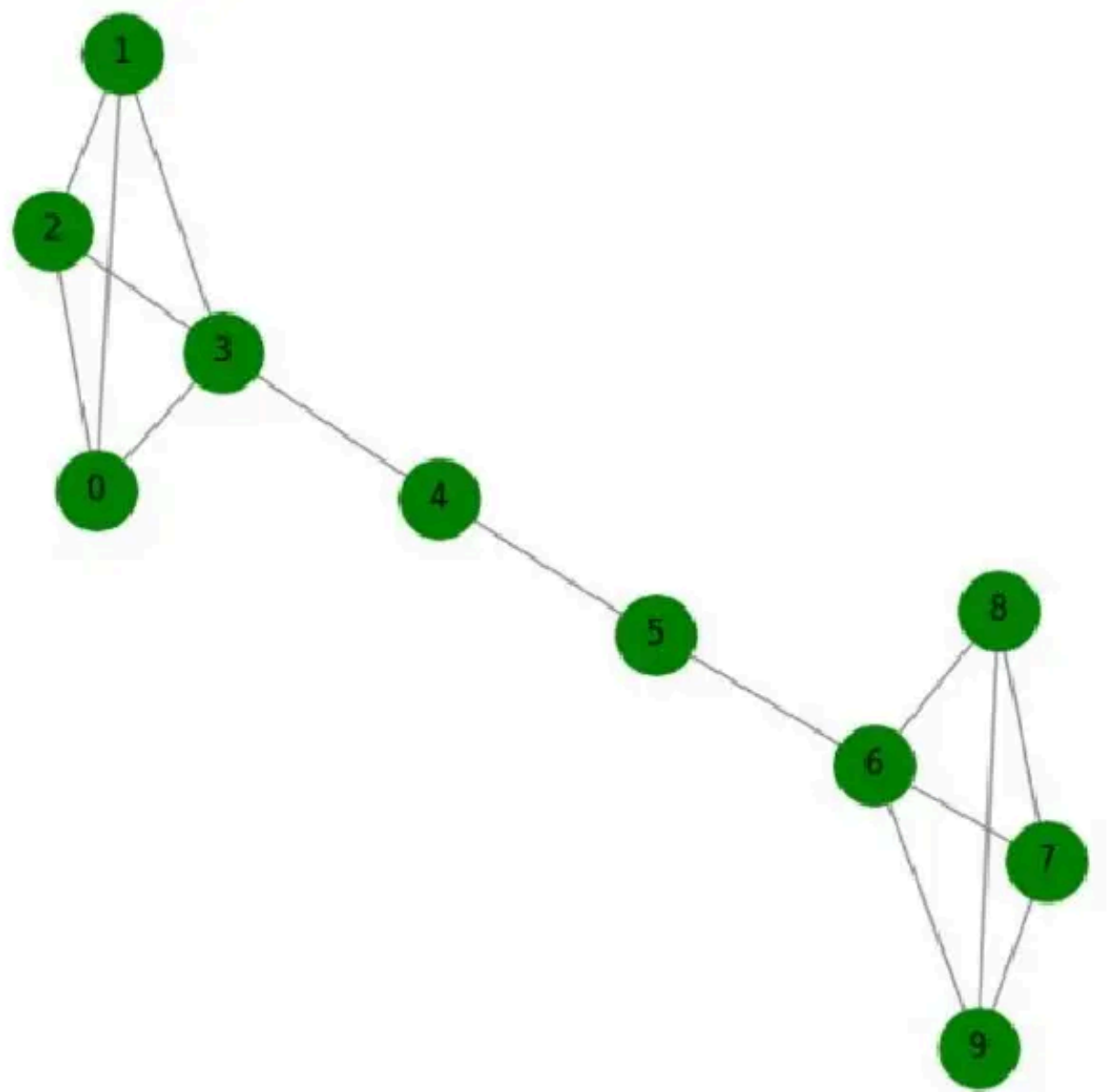
(assessment?
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Transcripts
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Books ()

Problem
Solving
Session -
Jan 2025 ()



- ☐ Ladder Graph
- ☐ Barbell Graph
- ☐ Wheel Graph
- ☐ Star Graph

No, the answer is incorrect.
Score: 0

Accepted Answers:
Barbell Graph

6) What will be the output of the following code?

1 point

```
import networkx as nx

G = nx.Graph()
G.add_edges_from([(1, 2), (2, 3), (3, 4), (4, 1)])
print(len(G.nodes()), len(G.edges()))
```

- ☐ (4, 4)
- ☐ (4, 8)
- ☐ (4, 2)

☐ (8, 4)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(4, 4)

7) What is the output of this code?

1 point

```
def collatz_steps(n):  
    steps = 0  
    while n != 1:  
        n = n // 2 if n % 2 == 0 else 3 * n + 1  
        steps += 1  
    return steps  
  
print(collatz_steps(7))
```

☐ 7

☐ 16

☐ 5

☐ 10

No, the answer is incorrect.

Score: 0

Accepted Answers:

16

8) What will be the first 10 numbers generated starting from $n = 7$?

1 point

```
def modified_collatz(n):  
    sequence = [n]  
    while len(sequence) < 10:  
        if n % 2 == 0:  
            n = n // 2  
        else:  
            n = 5 * n + 1  
        sequence.append(n)  
    print(sequence)  
  
modified_collatz(7)
```

- ☐ [7, 36, 18, 9, 46, 23, 116, 58, 29, 146]
- ☐ [7, 36, 18, 9, 28, 14, 7, 36, 18, 9]
- ☐ [7, 20, 10, 5, 26, 13, 66, 33, 166, 83]
- ☐ [7, 31, 15, 7, 36, 18, 9, 46, 23, 116]

No, the answer is incorrect.

Score: 0

Accepted Answers:

[7, 36, 18, 9, 46, 23, 116, 58, 29, 146]

9) Which technique is used in the PageRank algorithm to determine the rank of a web page? 1 point

- ☐ Depth-First Search
- ☐ Hyperlink Analysis
- ☐ Random Walk Simulation
- ☐ Data Scraping

No, the answer is incorrect.

Score: 0

Accepted Answers:

Random Walk Simulation

10) In the PageRank algorithm, what happens when a page receives multiple inbound links from highly ranked pages? 1 point

- ☐ Its rank decreases due to increased server load

- ☐ Its rank improves
- ☐ Its rank remains the same
- ☐ It is considered less relevant

No, the answer is incorrect.

Score: 0

Accepted Answers:

Its rank improves